

A Data-Driven and Principled Approach to Designing the Tokenomics of a New Blockchain-Based Game

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Abstract

With the advent of Bitcoin and the spectacular rise of crypto-assets, blockchain technology is now rapidly expanding to the video gaming industry. This evolution is the result of two converging trends, the “gamification of DeFi” (started by PancakeSwap) and the “financialization of games”. A new generation of blockchain-based games feature colorful characters and worlds that make both financial transactions more fun and the gameplay more stimulating. Active in-game trading of items and assets, an activity once reserved to expert players, is now commonplace in the open Web3 universe among casual or “midcore” types. On the other hand, compared to DeFi applications, which focus on a handful of financial services, modelling the whole economy of a Web3 game is a daunting task. In particular, its token economics (or “tokenomics”) involves all the on-chain circulation and complex interaction of game tokens, starting with their distribution among stakeholders (team, investors, and users) inscribed in smart contracts. To best prepare for this challenge, we conducted an extended analysis of established DeFi applications (Part 1), and current or forthcoming blockchain games (Part 2). This paper reports on our findings and how they guide the economic design of our own project, *Time Breachers* (Part 3). This new game consists of exploring, fighting, and managing resources in a complex “uchronic” multiverse composed of multiple versions of planet Earth. Given the richness of details offered by the game environment, one key decision was to rely on a single-token economy, where the game’s “share token” also plays the role of in-game currency. In part motivated by a survey that we conducted among dozens of “scholars” (full-time gamers), who strongly came out against juggling with two tokens, we also concluded that a unique token was a better choice to ensure an enjoyable, financially viable experience for users, and a profitable venture for token holders.

1 Tokenomics of DeFi Protocols

1.1 The Rise of DeFi

After the Bitcoin “big bang” of 2008, which simultaneously gave birth to the blockchain technology and the crypto universe we know today, the formation of the Ethereum galaxy in 2015 introduced smart contracts and became the prime home for most DeFi systems. One of the early pioneering projects was Maker, the creator of the first “stablecoin” \$DAI, rebranded as \$DAI later on, which was pegged to the US dollar and supported in the beginning only \$ETH as a collateral. In 2017, initial coin offerings (ICOs), a new mechanism to raise funds on-chain by minting native token denominations in exchange for \$ETH, were popping up everywhere. As this dense and overhyped ICO cloud eventually cleared, bright stars started appearing and expanding, some of which became major DeFi players who still dominate the space today: we can cite Aave, specialized in lending and borrowing (Section 1.4), and Synthetix, a liquidity application for derivatives.

From day one, the biggest challenge faced by all DeFi applications was finding clever ways to aggregate, retain and grow the indispensable liquidity reserves on which their entire edifice rests. In this domain, a major breakthrough came with the advent of Compound. Its idea was to incentivize users to lend and borrow by shelling out extra rewards in the form of \$COMP tokens. This quickly drove up return rates, which in turn fostered the practice of “yield farming”, whereby users were encouraged to go back and forth between lending/borrowing in order to maximize their gains. A new wave of projects followed suit, each one distributing tokens and opening more yield-farming avenues. Another groundbreaking concept put in practice by Compound was that of *governance*: owning \$COMP also gave users the right to vote on proposed modifications to the protocol (such as parameter tuning or rule changes)—a process constitutive of decentralized autonomous organizations (DAOs). This governance model is at the core of almost every DeFi and blockchain games today.

1.2 Main Characteristics

DeFi protocols fundamentally depart from centralized financial institutions in that they do not involve subjective component of risk evaluation such as “credit scores”, a rough indication of clients’ trustworthiness based on past behavior, which banks use to grant loans. Crypto-assets are managed automatically in a “trustless” fashion by smart contracts carved in the blockchain. A second major difference is that the various liquidity reserves used by the protocol to render services such as lending and exchange, are entirely provided by the users themselves (with perhaps the exception of its native tokens, see below), not created by central banks’ fiat.

Users’ deposits into the protocol’s smart contract accounts are typically acknowledged by issuing local “IOU” tokens, generally referred to as “LP tokens” for liquidity pool (e.g. \$aTokens in Aave, or \$xSUSHI tokens in SushiSwap). These receipt tokens generally bear the creditors’ interest by accruing value over time. They can also be traded on secondary markets before being eventually redeemed against the original deposits and burned.

1.3 Tokenomics Objectives

Beside the various LP tokens, DeFi applications first and foremost mint and manage their own token, called “governance” or “native” token. In many ways, a native token serves the same purposes as a firm’s shares in traditional finance. Both entitle their owners to a slice of the application/firm’s income and both give them a say in the governance through voting rights.

A fundamental difference between DeFi and TradFi, however, is the way the native token supply is distributed in crypto ecosystems. While firms rely on initial and seasonal public offering to attract capital, applications gradually disseminate their ownership rights to stakeholders and distribute a large part of the token supply to users. Moreover, instead of distributing dividends to shareholders only once a year, most protocols let investors stake their tokens in a “staking vault”, where revenues continuously accrue. To stress how fundamental this difference is, let us imagine a company like Amazon distributing a large part of its shares to its customers over the course of several years in order to bootstrap its adoption.

Across the board, and despite the great diversity of DeFi protocols, it appears that the many roles carried by their respective native token—which we call here “share token” (ST)—can be categorized into three main functions: funding, building network effects, and decentralizing the governance.

1.3.1 Funding

The primary use case of an ST is to raise funds through various funding mechanisms such as an initial coin offering (ICO) or an initial DEX offering (IDO) for decentralized exchanges. The team can also deal early investors (seed, series A, etc.) a portion of the future token supply. The ST serves in that case as an accounting mechanism to keep track of previous fundraising campaigns. Once the protocol is live and starts making profit, the amounts of ST distributed to early investors can be sold or staked to claim a prorata share of the profits.

1.3.2 Building network effects

Launching and scaling up new products from scratch by building rapid “network effects” remain a challenge. Given that a product’s value increases as more users engage with it, companies have to solve the “cold start” problem. This is especially true in crypto ecosystems, such as lending applications that simultaneously need many lenders and borrowers to be competitive, or decentralized exchanges that must expand by rapidly attracting both liquidity providers and traders. This is also vital in blockchain gaming, where a game’s value depends on how many players interact with it.

This is where the ST can serve as a means to create a network effect by rewarding specific actions that benefit the ecosystem, such as providing liquidity to targeted pools of an automated market maker (AMM), or borrowing in lending applications. If done right, aggressively distributing the protocol’s token to users proves an effective way to bootstrap growth.

1.3.3 Decentralizing the governance

The share token also helps govern the protocol in a decentralized way by gradually disseminating ownership among a large set of users, contributors and investors. Once the ownership is sufficiently spread so that no one holds a disproportionate share of the supply, governance rights can be activated. ST holders are then able to propose and vote on protocol changes, such as a modification of the fee structure, interest rate models, or other parameters.

1.4 Lending Applications

Lending protocols offer cryptoasset loans in a trustless and peer-to-peer (P2P) manner. Liquidity providers deposit their assets, which can then be borrowed by other investors. Defaulting from borrowers is normally prevented by requiring them to lock their own assets, or “collateral”, in greater value than the one borrowed. If the ratio of borrowed funds to deposited funds enters a risky zone, the latter are seized or “liquidated” to pay back the loan.

In most DeFi lending protocols, interest rates are programmatically adjusted by smart contracts as functions of the “utilization rate” (UR) of each liquidity pool, where UR is the ratio of total loans over total deposits in the pool’s currency. Gudgeon [10] propose a taxonomy of the interest rate models into three categories: linear, nonlinear, and “kinked” functions of UR. All models logically follow an increasing curve: the more a liquidity pool is utilized, the higher the rates become to incentivize users to bring back liquidity (lenders by taking advantage of higher profits; borrowers to avert higher costs). For example in Aave, the kinked function consists of two linear segments: first a slow rise for UR between 0 and some “optimal” value UR_{opt} (typically around 80%), then a steep slope shooting up from that point until 100%. The rationale behind this sudden increase is to strongly discourage more liquidity to be taken out when the pool is showing signs of drying up. The value of UR_{opt} and the linear coefficients are determined for each pool by Aave’s governing body after careful review of its underlying currency. Tokens deemed more “risky” are generally tuned at a lower UR_{opt} , typically 45%.

1.4.1 Aave

Over the past five years, Aave (“ghost” in Finnish) has become the undisputed leader of the DeFi lending/borrowing category. In 2017, the year of the ICO craze, it pioneered the field under the name EthLend. Its first version, however, offered exclusively P2P lending, i.e. users had to wait for an available counterparty to realize their loan request or offer. Meanwhile, Uniswap and Compound were making their entrance at the end of 2018 and introduced a new way to connect users indirectly via common pools of liquidity, each stored in a smart contract. Interestingly, this alternative peer-to-contract (P2C) model, which clearly facilitated transactions by suppressing the need for synchronicity among participants, also constituted a partial “recentralization” around hubs. It is important to note, however, that this took place in the virtual application layer sitting on top of the physically decentralized blockchain layer—which shows that the P2P architecture of the Ethereum node-based network does not need to be reproduced at the higher DApp level.

In 2018 EthLend decided to take the same P2C path and rebrand itself as Aave. Like the LP tokens of Compound called $\text{\$cTokens}$, depositors of liquidities into Aave pools receive $\text{\$aTokens}$ whose balance automatically increases with time by application of the various APRs. Liquidity providers can also use their deposits as collateral to borrow assets. In this case, as mentioned above, a maximum loan-to-value ratio r determines how much can be borrowed. If the fluctuating valuation of the deposit (continuously supplied by external “oracles”) goes down and brings r above the threshold, then any network participant can compete to liquidate the collateral.

On the other hand, Aave distinguishes itself by offering relatively “stable” (although not totally fixed) APYs to borrowers, and making an $\text{\$aToken}$ ’s value pegged at a 1:1 ratio to the value of the underlying supplied asset. Another fundamental mechanism of the Aave protocol is to encourage $\text{\$AAVE}$ holders to lock their tokens into the “Security Module”, a smart contract-based component playing the role of a mitigation tool in the event of a “shortfall” in one of Aave reserves. Thus, through a vote of the governors, the locked $\text{\$AAVE}$ can be used to compensate for the sudden occurrence of a deficit [2].

Overall, the added-value services that Aave protocol brings to the sector is a highly efficient and reliable non-custodial multi-pool liquidity market, giving users the possibility to access many currencies (34 as of today). It is also open source, meaning that anyone can build a third-party service or application to interact with the protocol and enrich one’s product.

1.5 Decentralized Exchanges

Another major DeFi instrument is swapping. Unlike centralized exchange platforms, such as Binance or Kraken, where the conversion pools are guaranteed and managed behind the stage by a single company, a **decentralized exchange** (DEX) protocol incentivizes its community of users to openly and collectively build liquidity via fees and other rewards in a crowdsourcing fashion similar to lending.

In general, a DEX follows two main algorithmic rules:

- Liquidity providers deposit amounts of paired assets forming a pool. Deposited amounts are equivalent in value. It means that in a pool $\text{\$A}/\text{\$B}$ where the exchange rate of $\text{\$A}$ is p $\text{\$B}$ tokens, providers depositing x $\text{\$A}$ must also deposit $y = px$ $\text{\$B}$ at the same time (e.g., twice for $p = 2$).
- Moreover, token valuations are adjusted after every swap in such a way that the product of the total quantity of $\text{\$A}$ contained in the pool multiplied by the total quantity of $\text{\$B}$ remains constant: $x.y = k$. Therefore, if a trader wants to convert $\text{\$A}$ ’s into $\text{\$B}$ ’s, the price of $\text{\$A}$ will increase, and that of $\text{\$B}$ decrease, after the transaction is completed (Fig. 1).

In exchange for depositing joint amounts of $\text{\$A}$ and $\text{\$B}$ in a pool, a DEX may grant liquidity providers pool-specific “A/B-LP fee-bearing tokens” proportionally to deposited amounts. This is very similar to the single-currency “A-LP tokens” emitted by lending applications.

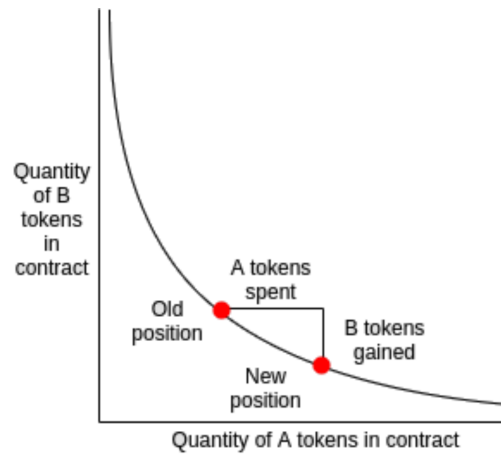


Figure 1: **Bonding curve.**

1.5.1 SushiSwap

One of the leaders of the DEX market today, SushiSwap, was originally forked from Uniswap. The number of paired-currencies liquidity pools that it offers is 354 at the time of writing. When it started in 2020, SushiSwap’s rather cavalier way of bootstrapping their protocol made headlines, as it consisted of encouraging liquidity providers to stake Uniswap LP tokens into SushiSwap’s own pools by rewarding them with \$SUSHI share tokens. The issuance rate of \$SUSHI was set initially very high, up to 1000 tokens per block, with a schedule planned to go down to 100 within a year. Given the high market valuation of \$SUSHI at the time, the interest rate of many of these pools exceeded 2,500% APY. The goal was to transform, or “migrate” Uniswap’s liquidities to SushiSwap—an aggressive strategy which was dubbed a “Vampire Attack”.

Today, SushiSwap rewards liquidity providers with fees collected from swaps, and with \$SUSHI tokens, as follows:

- Each conversion transaction incurs a 0.3% fee: it is composed of 0.25% added to the pool and 0.05% transferred to another contract, the “Sushi Bar” (see below).
- The Sushi LP tokens, called \$SLP, can be staked for additional revenue paid out in \$SUSHI, a process called *yield farming*.
- Furthermore, the earned \$SUSHI tokens can themselves be staked in the Sushi Bar in exchange for \$xSUSHI tokens.

The Sushi Bar is a special pool where \$SUSHI holders can stake their tokens and receive for them a certain amount of \$xSUSHI tokens, which is a special kind of LP token. \$xSUSHI accrues value thanks to the protocol fees mentioned above, as 0.05% of every swap is added to the Sushi Bar. At least once a day, a reward contract is called and all \$xSUSHI are sold for \$SUSHI. Staked \$SUSHI are shared among the shareholders of the Sushi Bar pool in proportion to their deposits. By design, 1\$xSUSHI should always have a greater value than 1 \$SUSHI.

To illustrate these rules with a toy example, let us assume that the staking pool contains a total of 1200 \$SUSHI brought by three stakers Rose, Sam, and Tom. In exchange, they received a total of 300.00 \$xSUSHI distributed as follows¹:

Stakers	\$SUSHI ¹	Share ^{1,2}	\$SUSHI ²	\$SUSHI ³	Share ³	\$xSUSHI
Rose	400	33.3%	$+400/3 = 533$	533	$\searrow 29.6\%$	100.00
Sam	200	16.7%	$+400/6 = 267$	267	$\searrow 14.8\%$	50.00
Tom	600	50.0%	$+400/2 = 800$	800	$\searrow 44.4\%$	150.00
+Uta ³	—	—	—	+200	$\nearrow 11.1\%$	+37.50
TOTAL	1200	100%	+400 = 1600	1800	100.0%	300 +37.50

At this point in time, the exchange ratio from \$SUSHI to \$xSUSHI was $1200/300 = 4$ for 1. Later, suppose that the Sushi Bar receives an extra 400 \$SUSHI coming from fees collected on swapping transactions². These gains are distributed among the current three stakeholders in proportion to their shares, which do not vary^{1,2}, and the new exchange rate is $1600/300 = 5.33$ \$SUSHI for 1 \$xSUSHI. Now, if another staker Uta deposits 200 \$SUSHI in the pool³, she will receive $200/5.33 = 37.50$ \$xSUSHI³. Because of the newcomer, previous stakers saw their shares of the pool reduced pro-rata, however their stakes were not diluted as they represent an unchanged quantity of \$SUSHI and the exchange rate remained at $1800/337.50 = 5.33$.

1.6 Tokenomics of the Share Token

The share token accrues protocol's revenue and/or provide holders with a say in the governance [7]. Most STs have both utilities although some are pure governance tokens (currently \$UNI of Uniswap and \$ENS of Ethereum Name Service) and others are only claims to the protocol's income (e.g. \$LQTY of Liquity). The genesis token distribution and emission schedule are the two pillars of a protocol's tokenomics. In the next section, we review their main features and implications.

1.6.1 Token distribution and emission schedule

Every new protocol starts with a pie chart which breaks down the token supply among (actual and future) investors, team and users. The distribution reflects a balance between the need for rewarding the investors and the team, on the one hand, and accelerating adoption by incentivizing usage, on the other hand. The genesis token allocation makes transparent to all parties the financial allocation and distribution over time, algorithmically enforced by the immutability of the genesis smart contract.

There are multiple ways to look at the supply distribution. One way is to focus on the circulating supply that excludes reserves and time-locked tokens. Another is to adopt the application's life-cycle view and examine at which rate the tokens are progressively emitted, and to whom. In the very long-term, only the maximum supply and fully diluted value matter.

The emission schedule is a key information beyond the allocation pie chart. Consider for instance the following two toy distributions:

	Distribution 1	Distribution 2
Investors/team, 1y locked	100m	100m
Community, 1y locked	100m	0
Community, 2y locked	0	100m
Community share	1/2	1/2

Both distributions allocate the same quantity of tokens to the investors/team and to the community, yet the second one is less favorable to the latter as investors and team members are allowed to sell their allocation before users are in position of doing the same. It is a simple financial principle that receiving an asset in the future is less valuable than receiving it in the present due to (i) the opportunity cost of not being able to obtain a return from investing the asset, (ii) the risk of selling at a lower price in case the protocol does not deliver on its promise, and (iii) losing the option of selling earlier rather than later.

Denoting $r > 0$ the period discount rate and $\gamma = 1/(1 + r) \in (0, 1)$ the discount factor, \$1 obtained in the last period is only worth $\gamma < 1$ now. In the previous example, while the investors/team allocation is worth $\gamma 100$, it is also $\gamma 100$ for the community in distribution 1 but only $\gamma^2 100$ in distribution 2. It follows that distribution 2 actually gives the community less than one half of the total supply in discounted terms.

In a more general setting, if we denote by a_t the allocation unlocked at date t to a particular group, and by s_t all unlocked tokens composing the circulating supply, then the discounted share of tokens distributed to this group reads:

$$\frac{\sum_{t=0}^T \gamma^t a_t}{\sum_{t=0}^T \gamma^t s_t}$$

where T is the number of periods after which all tokens are distributed. Protocols usually present the pie chart by assuming no discount ($r = 0$ or $\gamma = 1$), which obscures the fact that early recipients are favored with respect to late recipients.

As an illustration, we present the token distributions of two finance protocols, Euler Finance, a lending protocol and Curve an AMM protocol, both with and without flow discounting (Fig. 2).

The share distributed to the community (liquidity providers in Curve, borrowers in Euler) is higher in Curve than in Euler and the share given to investors is higher in Euler than in Curve. In both protocols, the share given to the team and investors is larger in discounted terms due to shorter vesting periods compared to the period over which incentives are distributed. This difference can be observed in the emission schedules of Euler (right-hand side graphic) and Curve (left-hand side graphic) in which distribution to investors is gradually unlocked before the distribution to the team and users (Fig. 3).

1.6.2 Token launch

A task that many applications fail to adequately manage is the token's price during the launch event. An initial small circulating supply coupled with a high token emission rate boost together the staking profitability and propel the price up. Several launches with an initial small float saw their token price reach unsustainable levels before crashing. As an

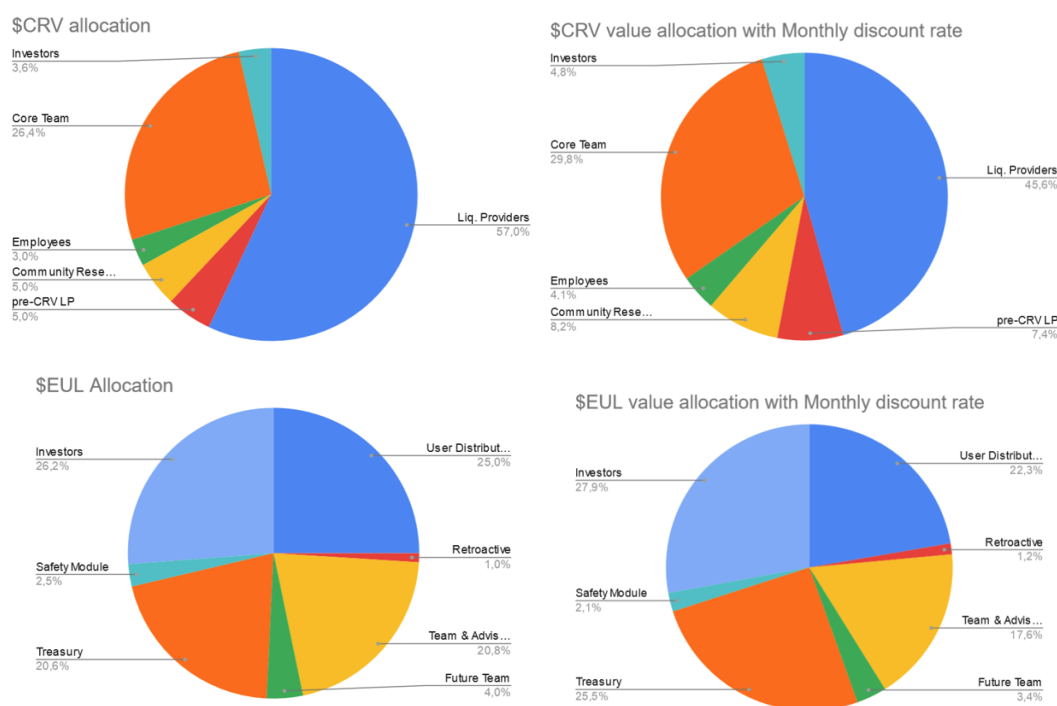


Figure 2: Curve and Euler allocation piecharts.

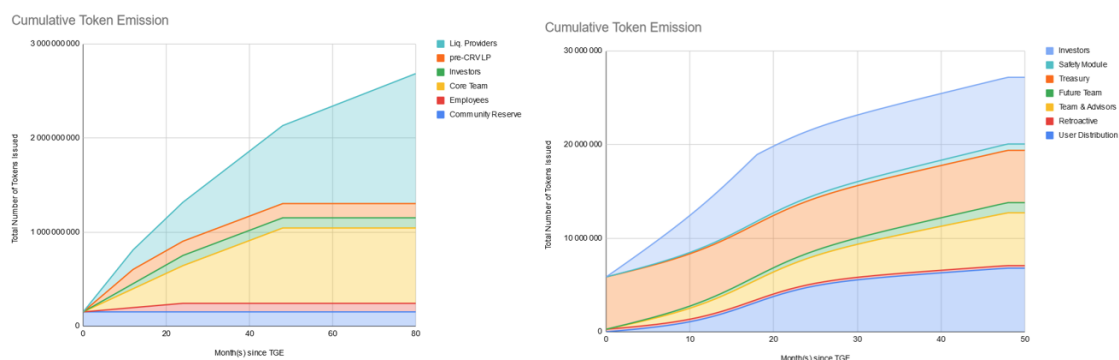


Figure 3: Curve and Euler emission schedules.

illustration, the price of Curve's \$CRV skyrocketed shortly after its launch on August 14, 2020 and the start of its liquidity mining program (Fig. 4). \$CRV started at US\$54 the first day (not visible on the Cointagecko graphic), then fell precipitously to around 50 cents. Investors attracted by high APR bought the token to profit from this abnormal yield.

A second example is given by the launch of \$LQTY, the token of Liquity (Fig. 5). Liquity provides \$LQTY holders with minting fees. Starting with zero float, the APR became absurdly high the first days and was arbitrated away by another skyrocketing price, which shot up from \$10 to \$100 in one day then plummeted back to around \$20 a few days later. A \$100 price briefly produced a fully diluted market cap of \$10b, which was more than four times the market cap of MakerDAO, Liquity's direct competitor.

The same problem arises for gaming protocols (which we review in Section 2). For instance, the game Star Atlas launched its share token \$POLIS on August 25, 2021 with less than 5% of its total supply initially in circulation. Immediately thereafter, another type of brisk up-and-down price dynamics could be observed (Fig. 6).



Figure 4: **Curve's \$CRV launch chart.**



Figure 5: **Liquity's \$LQTY launch chart.**

So why does the price climb so high at launch? To understand the mechanism, let us define a useful measure of financial performance: the price earning ratio (PER), which is derived by dividing the price p_t of the token by the protocol's earnings y_t per token:

$$\text{PER}_t = \frac{p_t}{y_t/s_t}$$

where s_t is the circulating supply at date t (and the total supply is normalized at 1). The PER indicates how much profit every token “buys”. At launch, the price of the token can be retrieved from the PER formula:

$$p_0 = \text{PER}_0 \frac{y_0}{s_0}$$

Since initial income is fixed and the PER is only mildly responding to the additional return provided by token emission, the initial price level is directly linked to the initial supply s_0 .



Figure 6: **StarAtlas' \$POLIS launch chart.**

This relationship means that a protocol starting with an initial float of 1% ($s_0 = 0.01$) of total supply will show an initial price approximately ten times greater than a protocol starting with an initial supply of 10%.

Price overshooting is obviously an undesirable event as many early buyers who believed in the protocol's value proposition are eventually led to sell at a loss. We propose two methods to mitigate the strong effect on the launching price: a sizeable initial supply and a lower emission rate at launch.

Launching a token with a large enough initial float can be done through various means: airdrop, public sale via an initial DEX offering (IDO) or an initial exchange offering (IEO, involving a centralized exchange), and unlocked tokens distributed to the investors/team. Classical public sales with a first-come-first-served rule are prone to "gas wars", however, which favor "whales" (big holders). To mitigate this risk, applications also use more innovative auction mechanisms.

For instance, Balancer AMM's "liquidity bootstrapping pool" (LBP) is an interesting solution for its ability to launch a token with low capital requirements. To achieve this, a two-token pool is set up with the project's token and a base asset. The weights are initially tuned in favor of the project token so that between 90 to 98% of the pool is made up of the project's token. As the weights are gradually reversed over time, a flip occurs where the base token constitutes most of the pool's value by the end of the token sale. The change occurs over a few days, facilitating price discovery. Another interesting avenue is "streamed auctions", an efficient and equitable price discovery mechanism proposed by the Locke Protocol (coming soon).

Another way of stabilizing the initial price is to start the distribution of the token for incentive purposes without allowing transferability between wallets. This way, the token has no market value in the early period until a decision of the DAO allows transferability. This is, for example, the solution adopted by the lending application Euler Finance (<https://www.euler.finance/>). The protocol could also make the token freely tradable but delay the distribution of its revenues to stakeholders until the circulating supply is large enough.

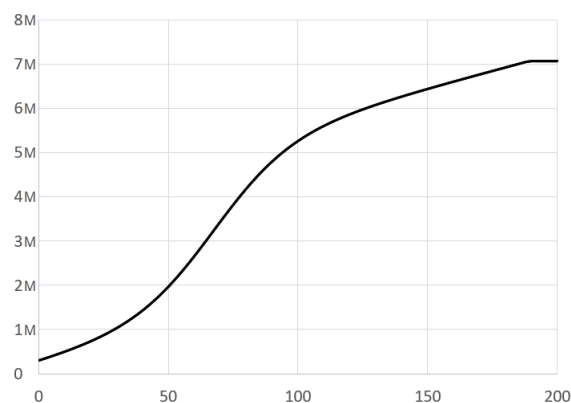


Figure 7: **S-shaped distribution schedule.**

A complementary approach is to optimize the emission rate. A high emission rate means that recipients are rewarded with a large share of tokens compared to the circulating supply. This translates into a high APR, which artificially pumps the price. The initial emission rate could be lowered by adopting a S-shaped distribution schedule. Interestingly, this option has been recently adopted by Euler Finance regarding the incentives in \$EUL given to liquidity providers (accounting for 25% of the total supply; Fig. 7).

1.6.3 Lockups

Many projects began experimenting with lockups for reward emissions in an attempt to reduce sell pressure and price volatility. The “ve-token” (vote escrowed) model invented by Curve Finance has been replicated by many protocols including Ribbon Finance, Frax Finance and Balancer Lab.

Liquidity providers stake their LP tokens on the DAO gauge to obtain \$CRV as a reward in addition to transaction fees. \$CRV rewards can be boosted by locking the \$CRV in exchange of a quantity of \$veCRV, which linearly depends on the amount of locked \$CRV and time lock expiry. Users can lock their \$CRV for a minimum of one week and a maximum of four years. \$veCRV holders can vote to channel \$CRV rewards towards one or multiple Curve pools. The ve-token model has proved successful in attracting and retaining investors willing to buy the share token. 55% of circulating \$CRV are currently locked with an average lock time of 3.6 years (Fig. 8).

Frax Finance, which also adopted the ve-token model, has 63% of its circulating supply locked. The ve-token model may create a virtuous cycle: by withdrawing a large portion of the supply from the market, the protocol benefits from a healthy price dynamics, which in turn attracts more liquidity by enhancing rewards for liquidity providers as well as more ve-token holders.

The ve-token model has also some downsides. Governance power can be centralized if a yield layer is built on top of it, and proposes staking derivatives to investors who want access to boosted yields but do not care about voting. For instance, Convex is currently the main holder of \$veCRV and possesses 46% of the voting power. A liquid staking derivative also means that holders can exit at any time, which defeats the purpose of lockups to align token holders with the protocol’s long term interest (Fig. 9).

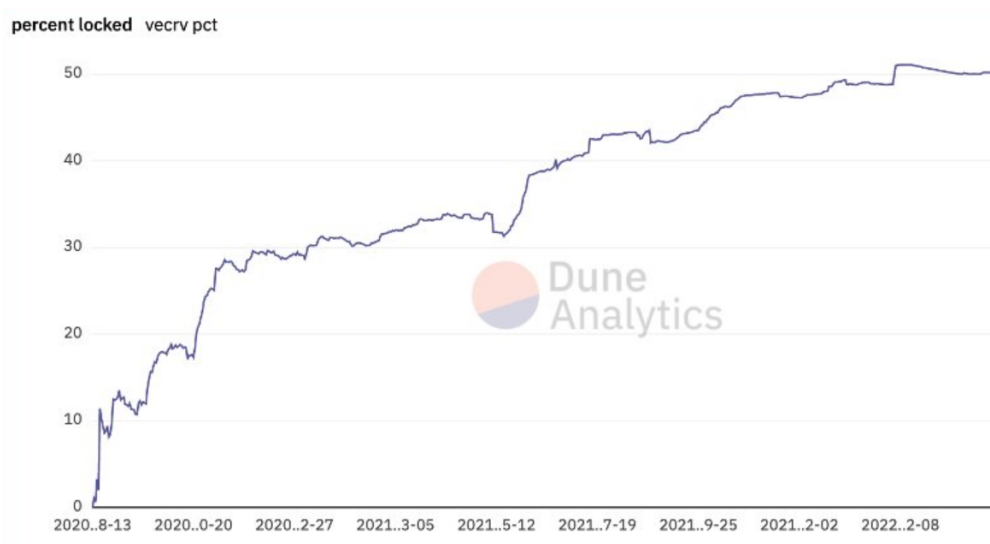


Figure 8: Locked \$CRV chart.

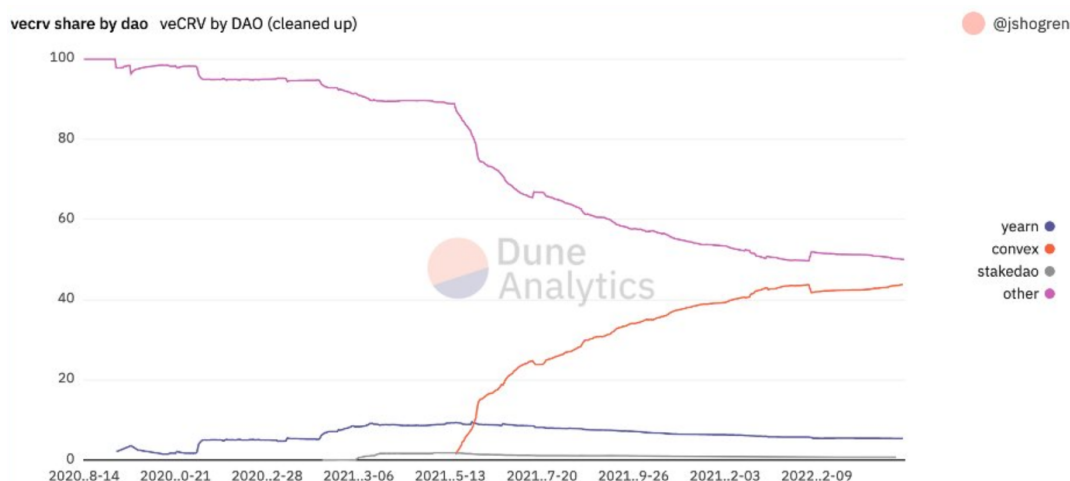


Figure 9: \$veCRV shares by DAO.

1.6.4 Liquidity mining

Liquidity mining refers to a protocol's practice of incentivizing user deposits with token rewards. This practice took off with Compound's introduction of liquidity mining in May 2020. The result on total value locked (TVL) was immediate and spectacular, with TVL going from \$100m to \$700m within a few days (Fig. 10). The number of daily new users jumped as a result and stayed at a high level one month later (Fig. 11). The success of the operation contributed to the 2020's "DeFi Summer" craze. Liquidity mining became a mandatory step for all DeFi protocols.

In recent months, however, liquidity mining has come under criticism. A recent study [12] shows that most of Compound's liquidity miners have little to no economic interest in the protocol and do not participate in governance. Liquidity incentives dilute token supply and add to selling pressure. According to the report, only 19% of the accounts have kept more than 1% of the \$COMP that they claimed, dumping 99% of their liquidity incentives on the market. A Messari report indicates that in Q3 2021, the Compound protocol paid out \$76m worth of tokens in liquidity incentives.



Figure 10: Total Value Locked in Compound.

Daily New Users – Compound

* by 2020-08-10

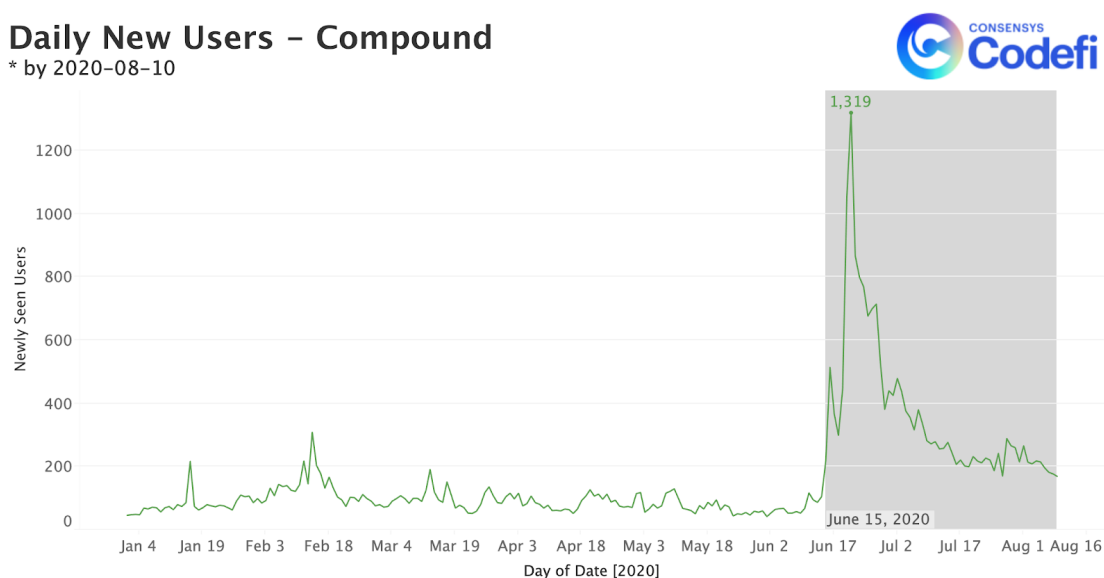


Figure 11: Daily New Users in Compound.

A possible improvement of the Compound model would be transitioning to a ve-token model as previously described, which would have the benefit of delaying the sell pressure by several years. Another path is to follow the approach pioneered by Olympus DAO. Instead of relying on providing incentives to liquidity providers, the protocol sells a portion of its token supply in exchange for a large quantity of base tokens (such as \$ETH on Ethereum), which is then deployed in association with another portion of the token supply to directly provide liquidity to DEXs. This way, the protocol does not need to incentivize liquidity providers to make its token liquid, while earning trading fees [18]. Stargate, a multi-chain bridge, has recently used this strategy with success to raise funds and provide liquidity to its own token.

2 Tokenomics of Games

A dozen years after the advent of Bitcoin and the spectacular rise of crypto-assets, blockchain technology started spreading to the video gaming industry where it is now rapidly gaining ground. The sector broke out in 2021 with the hypergrowth of Axie Infinity. Since then, the Web3 gaming market has expanded exponentially, capturing a sizable share of the gaming industry.

This evolution is the result of simultaneous and converging trends. On the one hand, the “gamification of DeFi” initiated by PancakeSwap (taking after the model of Robinhood in traditional finance) created a fanciful dressing around arduous mathematics to attract a younger and/or less technologically inclined audience. A user-friendly and colorful interface featuring fun characters and a backdrop story is more likely to be adopted by users already at ease with gaming and social media. On the other hand, the “financialization of games” (which previously existed with “gold farming” in massively multiplayer online (MMO) games), was taken to a whole new level by the advent of cryptoassets, as exemplified by Axie Infinity. The trade of assets, skins, cosmetics, and other resources, an expert activity once reserved to dedicated streamers and gamers inside the proprietary world of a game studio, is becoming commonplace in the open Web3 universe.

This new alliance of gaming and DeFi combines playful elements such as avatars, creatures, battles, rewards, accessories, land, and so on, with financial primitives such as lending, borrowing, trading, staking, NFT minting, or yield farming. It makes both the investment services more fun and the game process more captivating and challenging. Generally dubbed “play-to-earn” (P2E) today, in contrast to the legacy “pay-to-play” model, the purpose of blockchain games is to offer economic incentives to grow and retain large communities of players without requiring an entrance ticket. Typically, players can earn fungible and nonfungible tokens from scratch by completing tasks, winning competitions with other players, and progressing up the game levels. A new possibility offered by most Web3 games is also to let players transfer their pot of money and trophies out of the game’s world to NFT marketplaces and crypto exchanges. A DAO in which players govern as a community the evolution of the game is also a core objective of natively web3 games [16].

For this study, we reviewed four different Web3 games at different stages of production and release, namely: Axie Infinity, Illuvium, Star Atlas, and Aurory, especially focusing on their token economics. We also looked at a class of protocols more akin to gamified DeFi than actual gaming, such as DeFi Kingdoms.

2.1 Objectives of a Game Tokenomics

The term tokenomics refers to the on-chain mechanisms (and sometimes legal terms) that (a) govern the way tokens are minted, earned, distributed, and possibly burned; and (b) confer utility to the token holders. By rewarding a set of actions it contributes to the system of incentives which encourages desirable and constructive behaviors in the ecosystem [1].

Designing the whole tokenomics of a blockchain-based game (BBG) is a daunting task [17]. Compared to DeFi protocols, which focus on a single financial service, modern

BBGs aim to recreate a full-fledged monetary economy equipped with a comprehensive set of financial primitives that enable participants to trade, save, lend, and borrow. Those primitives support a game ecosystem populated with a variety of tokens, fungible and non-fungible, which represent all sorts of in-game assets and resources. Those assets are priced according to complex and intertwined market forces. A rich gaming and financial environment populated by hundreds of thousands, sometimes millions, of players interacting with each other make the whole system nearly as complex as what can be observed in real-world societies.

This is why a necessary first step toward a well-designed tokenomics is to build a *model* of the main interactions among agents and/or macro-level quantities in order to gain a better understanding of the in-game market dynamics. Only by operating within a healthy ecosystem can the game deploy its full potential and scale up. On the contrary, a flawed design can quickly lead to a dead end, such as an inflationary economy with devalued game assets and players leaving en masse. But if done properly, a solid tokenomics can give the game's creators a great competitive edge in a highly promising and profitable sector.

Modelling the complex structure of a game tokenomics can be often clarified by distinguishing upfront between two types of tokens: a *share token* (ST), which accrues profits from game usage; and an *in-game currency* (IGC), which bears utility for the purpose of the gameplay itself. Following this classification, many gaming platforms, such as Axie Infinity or Illuvium, have adopted a “dual tokenomics”, in which these two types of tokens coexist and are ascribed distinct objectives. We review them in the next sections.

2.2 Share Token

Similar to regular DeFi (Section 1.1), an ST represents ownership rights to incomes generated by the game. To avoid uncontrolled dilution, it usually comes in limited supply and with a predetermined emission schedule set in stone from day one in a genesis smart contract. Then, the circulating supply increases over time to match what is hoped to be a growing game adoption. In sum, an ST essentially serves the same three DeFi functions reviewed in Section 1.1: funding, liquidity incentives, and decentralized governance—with a few qualifications.

2.2.1 Funding

Token emission is the main vector by which the team and investors are paid for their contributions to the game's existence. The tokenomics must be structured so that the ST holders capture most of the value of the game. This is classically achieved by providing token holders with ownership rights to the staking vault, into which fees collected during the game are deposited.

2.2.2 Liquidity incentives

A DEX liquidity pool, where the ST is paired with the blockchain's native token (e.g. \$ETH in Ethereum or \$SOL in Solana), is a critical infrastructure of any blockchain-based protocol, including gaming applications. A deep liquidity pool allows participants to buy and sell large quantities of ST without major price impact. By enabling a two-way convert-

ibility, the pool serves as a bridge between the protocol and its blockchain ecosystem, over which values can be transferred back and forth.

Providing liquidity in this type of pool is extremely rewarding if the protocol takes off. It is also a high-risk endeavor in a start-up project where losses can be amplified by the mechanics of automated market makers (AMM), on which liquidity pools rely. Given both the importance of the liquidity pool for the protocol and the high risk for liquidity providers, most protocols heavily incentivize the pool by distributing a sizeable portion of their emitted ST. In exchange for subsidizing the pool, most games have adopted a locking system for LP tokens with the objectives of temporarily removing selling pressure from the market and aligning stakers' incentives with the game's long-term interests [13].

In **Illuvium**, stakers have the option of locking their LP tokens on a sliding scale for any period of time up to 12 months. Those who choose to keep their staked tokens unlocked have the flexibility to withdraw them at any time. Those who lock them gain a boost that linearly increases with lock duration. Locked staked tokens are weighted up to twice the unlocked staked tokens.

As a side effect, the liquidity-mining program helps decentralizing further the distribution of the tokens among the gaming community.

2.2.3 Governance decentralization

Compared to centrally controlled traditional games, BBGs have the potential to remove all middlemen and gradually transition to a fully decentralized application owned by its users and capable of running quasi autonomously. Initially concentrated in the hands of a small number of stakeholders, essentially the team and investors, token ownership and associated governance rights are gradually distributed more evenly across participants. Platforms with higher degrees of decentralization provide better protection against censorship resistance and discourage bad actors from manipulating the governance or skewing the game to their advantage. Distribution is a long-term undertaking, however, which can require many years before achieving full decentralized ownership and governance.

Axie Infinity, which pioneered the era of P2E games, is leading the way toward full governance decentralization. In 2021, its "Community Treasury" began to receive revenues generated by the game as well as a portion of staking rewards. This treasury will be entirely governed by \$AXS holders in 2023, once the network has reached a sufficient degree of decentralization.

Other gaming platforms reviewed in this survey are still at an early stage but envision a similar long-term path.

2.3 In-Game Tokens

In contrast with ST, in-game tokens encompass cryptoassets which are used in the game and whose ownership is secured by the blockchain. Three types of assets can be distinguished: in-game currency, NFTs, and various other fungible resources.

2.3.1 In-game currency

While the economic concept best suited to grasping the logic of share tokens are a firm's shares, the closest equivalent of an in-game currency (IGC) is fiat money. Share tokens are expected to reward investors by a price trending up, whereas an IGC is expected to maintain a stable value in the game in order to facilitate exchanges among other in-game assets like NFTs and fungible resources. The mass of GT should roughly increase with the number of players and the game's volume of activity. Contrary to the ST, its price should not automatically rise with playing activity. Another difference is that the IGC may or may not have a tradeable market value.

Illuvium relies on a dual token economics with an ST (\$ILV) and an IGC (\$sILV), yet \$sILV does not have a market value. It cannot be bought in an exchange but instead is distributed through the staking vault. Players who stake their \$ETH/\$ILV LP tokens in the vault can choose to be rewarded in either 12-month locked \$ILV or \$sILV. In the latter case, \$sILVs are minted on the fly and readily available in Immutable X (\$IMX) tokens, which can be traded for in-game NFTs or fungible resources.

When the IGC has a market value, it is preferably paired with a stablecoin in a liquidity pool. Insofar as an IGC maintains a stable value in US dollars, this pairing choice minimizes the risk of impermanent loss for liquidity providers.

Similar to the monetary mass in real world economies, the total emission is either not capped, or capped at an upgradable level. To provide players with in-game rewards, the tokens are generally minted/burned on the fly, as in Axie Infinity or Illuvium. Sometimes, the IGC obeys a fixed total supply and is distributed like the governance token ST to investors and other stakeholders, as in Star Atlas. Instead of being minted on demand, it is withdrawn from the Treasury, which itself receives a continuous flow from the genesis contract (as in Star Atlas).

2.3.2 NFTs

NFTs are certificates of ownership of a digital good secured by the blockchain. They play a key role in BBGs as they make players owners of their in-game assets, such as skins, characters, weapons, or virtual lands. The scarcity of in-game NFTs is provable through the immutable records embedded in the blockchain network's smart contracts. Character NFTs provide players with a unique visual identity, which facilitates their immersion in the game's universe. Not all items within a game should be NFTs but only the most important assets whose characteristics are unique and can be upgraded.

Many games have expanded the utility of NFTs by attaching special perks to them. In **Aurory**, NFT holders or "Aurorians" are granted early access to future game modes and land sales. They will also enjoy better rewards than free-to-play (F2P) players. In **Illuvium**, players can customize their NFT Illuvials. Each Illuvial has a class and an affinity and can cooperate in battles with greater force when their team mates have the same classes or affinities.

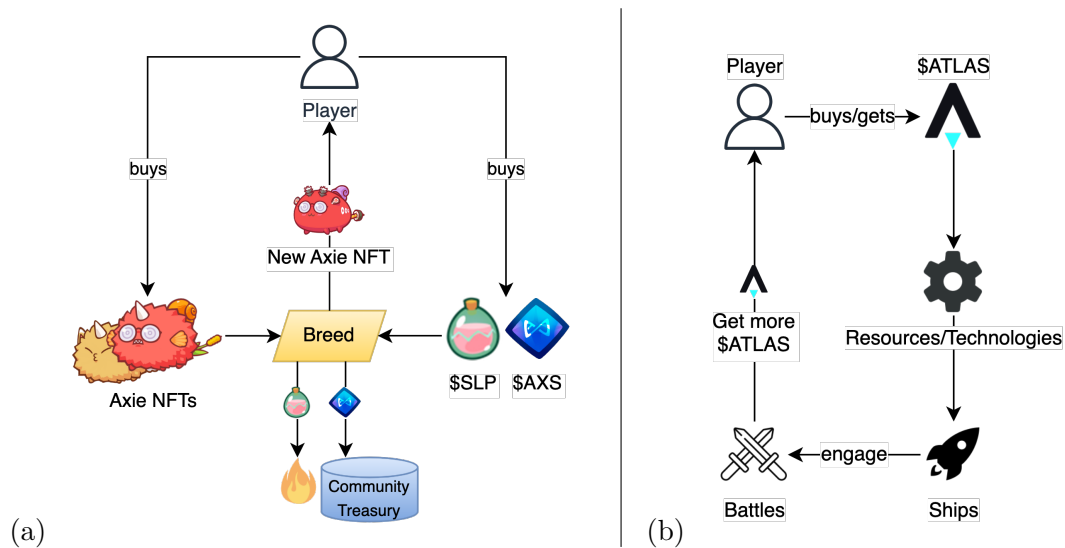


Figure 12: Axie's and Star Atlas' core loops.

NFT indivisibility may be an issue, however, if their price dynamics requires a significant upfront investment prohibiting more players from participating, as illustrated by the lack of accessibility of Axies during price booms.

2.4 Modelling a Game's Tokenomics

As stressed in the introduction, designing the tokenomics of a game can be challenging as it necessitates a good grasp of the complex interplay between the various tokens and assets used in the game. Modelling the game's economy is an effective way of tackling this complexity. A model is a simplification of the game's economy that concentrates on a few properties relevant to the analysis, while abstracting away non-essential features. A model helps understand the major forces at play, and possibly predict their outcomes. Two complementary approaches to modelling a game's economy focus on "reward loops" for one, and "source/sink flows" for the other.

2.4.1 Reward loops

Reward loops are simplified representations of gaming strategies that give players a way to earn valuable in-game resources. They constitute strong incentives for users to engage and progress in the game. At the most basic level, a reward loop starts with some amount of resources, goes through several stages and ends with a possible increased amount of resources. Identifying the reward loops allows to assess the financial utility of each token in the game. Its utility, and eventually its price, will be determined by its capacity to unlock the rewards.

For example, in Axie Infinity, a core loop rewards players with new Axies [11] (Fig. 12a). Users enter the game buying Axies, \$SLP and \$AXS tokens with actual US dollars. Then, they engage in breeding to generate more Axies. Here we can see that the two main tokens and the NFTs have to be combined in certain proportions to beget more Axies. A consequence is that, everything else being equal, their price should be positively correlated since more breeding raises the demand for all tokens simultaneously [3].

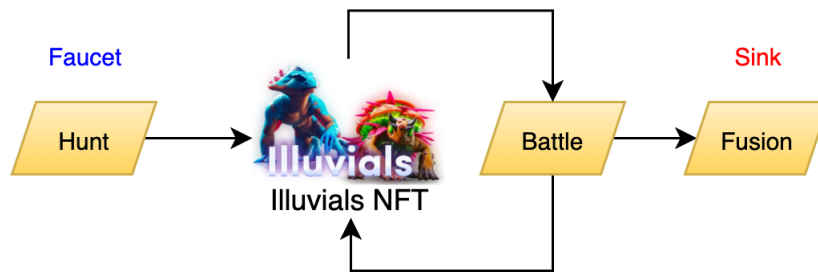


Figure 13: Illuvials.

In Star Atlas, players use the IGC \$ATLAS at various stages of the game to mine and refine resources, build innovative technologies and eventually craft a starship. Once equipped, the ship can be engaged in battles to earn more \$ATLAS (Fig. 12b). In this example as in the previous one, P2E consists in combining the IGC with other in-game NFTs and resources to earn additional IGC or NFTs. The reward circuit involves several resources at different phases of the game. Extensive simulations under various scenarios should check that the expected gain is both positive and sustainable.

2.4.2 Sources and sinks

Sources and sinks refer to the various ways a token enters and exits the game. Examples of sinks are paying for utilities, rare collectibles, gambles, crafting, upgrading, boosting, breeding, items that confer status, shortcuts that save time, and fees [15]. Identifying and quantifying the various sources and sinks of a token allow to evaluate the rate at which its circulating supply is building up and, ultimately, the actions that mitigate selling pressure [5].

Some in-game tokens exhibit a natural balance between sources and sinks. In Star Atlas for instance, minerals are mined and refined by players, then absorbed in the production of ships and equipments. Other tokens have only sources and for this reason are deemed inflationary, at least in the long-term. An example is given by the Axies, which are minted through breeding but cannot be burned. Despite a surge in price during periods of frenzy, the demand for Axies cannot indefinitely keep up with its growth.

In Illuvium, players can “merge” their Illuvials, once they reach a certain level, and create new ones with greater power. This possibility adds a useful sink and mitigates the risk of price depreciation over time (Fig. 13).

Balancing sources and sinks is especially critical for an IGC. Its mechanism of emission and destruction should be designed so that it maintains a relatively stable secondary market price. In Axie Infinity, the production of \$SLP is uncapped. The lack of \$SLP consumption outside of investment demand other means by which its price should inevitably trend toward zero. Therefore, the challenge is to find new utilities for the IGC by making the game more complex or proposing non-monetary uses. In accordance with this logic, Axie’s team is planning to add a sink in the next version of the game via the possibility of buying cosmetics (Fig. 14).

The emission of the IGC can also be regulated by a Decentralized Autonomous Organization (DAO) governed by the ST holders, the DeFi equivalent of a central bank.

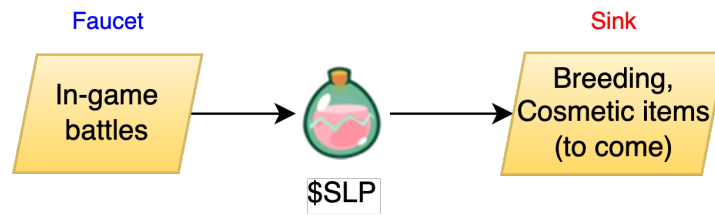


Figure 14: Axie's \$SLP.

In **Star Atlas**, the future DAO will adjust the emission rates of \$ATLAS by tuning the multipliers applied to in-game rewards. The targeted price is a compromise between making the game accessible to a broad set of players, on the one hand, and providing a reasonable revenue for the ST holders, on the other hand. In the long term, this discretionary system will be replaced with an automated stabilization algorithmic, with the aim of decentralizing the game further.

2.5 Single-Token vs Two-Token Economies

Most BBGs have chosen a dual-token model with a separation of the ST and IGC. Thanks to the clear conceptual distinction between the gaming and investing functions, each token serves a well defined purpose (see Appendix A for a formal approach). The IGC maintains a stable price by balancing the buying pressure exerted by new players entering the game with the selling pressure from existing players cashing out in-game rewards. The ST price increases with the growing number of players through the fees collected in the game. Advocates of a two-token game economy outlines that merging the two assets may expose the game to two opposite dangers: players with little wealth are excluded from the game if the single token becomes too expensive to acquire (lock-out risk); or conversely, in-game rewards could exert an inflationary pressure on the token market and prevent its price from increasing.

Despite the benefits of the dual-token model, some games have opted for a single-token model in which part of the ST supply is distributed to players as in-game incentives.

In **Aurory**, a unique token \$AURY, featuring a fixed supply, serves both in-game and investment purposes. However the game's designers took care of mitigating the effects of a possible price increase on the game's accessibility by adding two non-tokenized sub-currencies: OKA, earned by financially invested players, and TOKE, earned by F2P players. These soft currencies are collected in-game and exchanged each week for a quantity of \$AURY according to a predefined formula. This way, both invested and F2P players can participate and progress in the game whatever the price of \$AURY may be.

One potential drawback of the single-token model is that in-game incentives depend on the distribution of a token whose emission is limited in time. Moreover, even though the in-game distribution circumvents the lock-out risk, the inflationary risk remains.

The game **Defi Kingdom** on the Avalanche subnet also adopted the single-token model with its ST \$JEWEL, which is distributed for staking, yield farming and during quests. To counteract inflationary pressures, those sources are balanced with multiple sinks such as “summoning” (minting) new heroes, a marketplace, and yield-farming withdrawal penalties. An new type of sink takes the form of gas fees paid for every in-game transaction not collected by the blockchain network but kept by the game.

On the other hand, the single token model also presents some advantages. Distributing the main token as in-game rewards avoids the segregation of players and investors, and contributes to building a cohesive community. Moreover, the multiple uses of the main token strengthen its overall utility and market value as shown in Appendix A.

2.6 Lessons Learned

Following in the footsteps of the now “classical” model adopted by hundreds of DeFi protocols over the past few years, the task of designing an ST emission greatly benefits from the battle-tested templates they produced. In essence, they all feature a fixed supply, a transparent distribution, an algorithmic emission schedule, and various locking and vesting mechanisms.

By contrast, compared to the share token, the design of an in-game token is still an open research field. The goal of the IGC should be to ease a continuous and sustained “onboarding” of players, and this hinges on an affordable price and a healthy price trajectory to prevent downward price expectations. Pursuing this line of reasoning, we believe that the current two-token systems initiated by Axie and followed by Star Atlas and Illuvium have not demonstrated their superiority given the downward price dynamics recently observed.

Therefore, it is time to innovate and propose an alternative tokenomics model based on a *single token* fulfilling the functions of an in-game means of payment, a share token and a governance vector. The single-token system is better suited to the model adopted by *Time Breachers*, in which a large part of the game is free access. This unique token will unify the player community, and help onboard new player profiles through the game’s incentive system and distribution of the token. Those lines of arguments are developed in the next section devoted to the tokenomics.

3 Tokenomics of *Time Breachers*

3.1 Game Description

The game devised by Starchain Gazer, *Time Breachers*, belongs to the “4X” genre, which stands for “eXplore, eXpand, eXploit and eXterminate”. It refers to a type of strategy game in which the player controls and manages a whole empire. Known for their depth and level of detail, 4X games focus on the economic and technological development of a virtual civilization that the gamer must create and lead to supremacy. They usually involve a complex gameplay taking a great amount of time to maneuver and complete. To assist in this endeavor, a lot of the micromanagement tasks can also be automated.

In this context, the essential activity of *Time Breachers* consists of resource management in a complex, combinatorial and “uchronic” multiverse. Set in space and on land, it features a potentially infinite number of different versions of planet Earth, offering many bifurcating paths of progression and gains. Players travel aboard a Mothership and through time between these alternate planets, each one characterized by a unique mix of up to three different “eras” picked from a wide variety of natural and/or cultural environments, such as: dinosaurs, Vikings, Middle Ages, WWI, nanorobots, Matrix-like, and so on. Note that, despite the great degree of imagination involved here, the game does not qualify as plain “fantasy” since no magic or supernatural elements are involved. Displaying at most science-fiction-grade advanced technology, within improbable but not impossible parallel worlds (e.g. one version of Earth invaded by aliens), it can be deemed “semi-realistic”.

The core task of a player is to create a squad of warriors, also called “Uchronians” or simply “Units”. True to the eclectic spirit of the game, Units can exhibit up to three contrasting cultural and historical traits in a fun and unexpected way, e.g. Napoleonic soldiers with ray guns, astronauts riding T-Rex’s, and so on. At the core of the game’s business, users can trade or loan their Units and accessories, even rent out their Mothership, to other users.

3.1.1 Game phases

In accordance with the 4X nature of the game, players go through different phases in turn, which may overlap. First comes *exploration*, in which the Mothership moves from planet to planet usually every few days, depending on the user’s preference. Each planet hosts a cluster of occupants or “Time Pioneers”, representing a maximum of 600 concurrent players. It harbors a unique combination of climate, geology (tropical forest, desert, volcanoes, ice, etc.), creatures, technology, and two “eras” inspired by Earth history (Roman empire, Victorian England, space age, etc.). A slew of events and interactions take place at this point. Typically, an initial “fog of war” surrounding the gamer’s Units gradually dissipates as they move and discover their local environment (trees, rocks, other characters). During exploration, players consume resources and use utility items, in particular “probes” and sensors to gain more visibility, or missiles to prevent other players from using their own probes.

When a team of Units reaches a “Time Breach”, a kind of spatiotemporal portal, it is teleported to a planet’s surface where it is immediately thrown into battle. This event kicks off an *extermination* phase, a gameplay similar to that of classical strategy games such as *Age of Empires*. Here the objective is moving a squad of typically three to five characters from point A to point B, where they must defeat enemies in order to earn assets called “Time Cubes” used to create new Artifacts (accessories, see below) and Units. As soon as a team steps through a Time Breach, it is randomly dispatched to a completely different location harboring surprising environments and enemies. Since Units fight with their own competence and items depending on their epoch or origin (dinosaur, laser gun, coat of mail, etc.), the outcome of the battles can vary greatly. Whenever an enemy falls, Time Cubes appear in its place. Then, three Artifacts (one weapon, one armor, one utility item) give rise to one new Unit, contributing to the *exploitation* phase.

Resource crafting can also bring precious technical upgrades to the Mothership, enhancing its sturdiness and ability to travel to harsher or extreme environments (e.g. volcanic or icy planets), thereby enabling the *expansion* phase. Note that, while the Mothership is the only geographical invariant of the players, becoming more specialized in a subset of environments should be a better strategy on the long run than remaining general-purpose. The game settings should lead players in that direction.

3.1.2 Game usability

The game strives to be accessible and engaging for a wide variety of users and personalities. One typology concerns the frequency and commitment of the player and involves four categories:

- casual: 10-20 minutes 1-2 times per day
- midcore: 20-40 minutes 2-3 times per day
- hardcore: several hours per day, or until all current resources are used
- native Web3 gamers, or “Web3 native” in short: blockchain-knowledgeable individuals spending most of their time playing and re-investing assets acquired during the game; they also target leaderboard prizes

On this basis, the game’s breadth of appeal should make it equally easy to just briefly visit or to explore and exploit it in depth.

Another classification looks at a player’s psychological profile or behavioral preferences: it distinguishes the “killer” (mostly interested in fighting opponents), the “achiever” (focusing on long-term objectives and thorough progression), the “explorer” (curious to discover new environments), and the “socializer” (essentially there to interact and chat with other players). Here too, *Time Breachers* should welcome all profiles, which potentially amounts to 16 types when independently crossed with the four levels of commitment above—although some combinations are obviously less plausible than others, e.g. hypothetical “casual achievers” or “socializing Web3 natives”.

3.2 Game Assets

Due to its complexity, *Time Breachers* features many quantities and assets of various types. Some are stored off-chain, such as: Energy Points, Time Fluid, Experience Points, and Governance Score; others translate to on-chain tokens, such as: Time Stones, Time Cubes, Artifacts, Units, and Entry Tickets.

3.2.1 Off-chain quantities

Energy Points As in most games, Energy Points (EPs) are a unit of measure allowing players to participate in certain activities by controlling their progression and session duration. The amount of EPs required for each challenge (travelling, fighting, and so on) is related to the difficulty of the challenge and its corresponding rewards. EPs replenish naturally over time, up to a maximum amount that also increases with levels, or they can be purchased via special “boosters” on the marketplace. EPs are nominal and cannot be traded or shared among players.

Time Fluid Here, Time Fluid (TF) is a “soft currency” used to purchase diverse items or pay costs and fees. It is earned via engaging in different gameplays such as Time Breaches during an action phase (in limited quantity, about 5%), or Time Veins for asynchronous offline mining (to a much larger extent, around 80%). TF can also be purchased in TF “packs” on the marketplace. The maximum quantity of TF per player is not capped.

Experience Points Experience Points (XPs), also called “XP Dust”, are an important resource to level up Artifacts. Whereas the highest quantity of Time Fluid is earned in Time Veins, the main way to gather XP Dust is to raid Time Breaches. XP Dust is distributed to all the Units of a player. Players can also purchase or earn XP Dust boosters to improve their standing. Specific Battlepasses (see below) and “social privileges” (belonging to a clan or a guild) can also boost XP gains. XP Dust may or may not be capped.

Governance Score All players have a Governance Score (GS) that allows them to vote, make decisions or obtain privileges within the game. The GS represents a skill level (similar to Elo-MMR systems) composed at 60% of all kinds of player activities and degrees of engagement in the game (regularity, session time, purchases, craft, challenges, DeFi, socializing, and so on), and at 40% of token-staking income issued by a dedicated Governance Staking Pool. The GS is “degenerative”, i.e. players can lose portions of their scores depending on their choice of activities or lack thereof. Thus, keeping a good GS requires a regular maintenance routine and recurrent engagement with the different activities of the game.

3.2.2 On-chain tokens

Time Stones Time Stones (\$TSTs) are the only fungible token of the game, in compliance with the ERC-20 standard, and represent its main economic currency (Section 3.4). \$TSTs are used to mint or upgrade both Artifacts and Units, and to purchase other items on the marketplace. The main way to collect \$TSTs is to complete Battlepass quests (see below), including various competitive multi-player challenges (PvP, PvPvE, Hyper-Breaches). Another way is to buy \$TSTs on the marketplace. In general, an Entry Ticket is required to access the most \$TST-rewarding challenges. The maximum amount of \$TSTs that one player can own is not capped (within the total quantity available).

Time Cubes As mentioned above, Time Cubes (TCs) are elements that must be combined to craft Artifacts. Contrary to Time Fluid, Time Cubes are mainly obtained in Time Breaches (around 80%) while also found in smaller quantities in Time Veins (around 10%). The rest is generated by various actions. All TCs possess the same characteristics. Generally about 10 TCs are needed to create an Artifact, but players can increase their odds of obtaining an Artifact of greater rarity by using more TCs. The maximum number of TCs collected by a player may or may not be capped depending on the Mothership’s storage facility.

Artifacts Artifacts are crafted in a dedicated space called the “Summoning Chamber”, by combining Time Stones, Time Fluid and Time Cubes. The more resources are used, the better the odds of obtaining higher-quality Artifacts. Occasionally, Artifacts can also be earned via specific activities such as Time Breaches or PVP rewards. Artifacts minted with Time Stones can be directly purchased on the marketplace. There are three types of Artifacts: Weapons, Armors, and Utility Items, each characterized by a rarity level and an era. In general, a newly crafted Artifact is not immediately registered as an NFT, but minted Artifacts are. An Artifact features innate statistics and a notion of “skill”, which progresses as the Artifact gains more Experience. This progression is randomized for statistics, while becoming faster for rarer Artifacts.

Units Units are just another name for the main characters of the game, the Uchronians. They participate in different gameplays as warriors or miners. A Unit is crafted, or “summoned”, in the same Chamber as above using three Artifacts (one weapon, one armor, and one utility item). Units inherit from the combination of rarity and eras of the Artifacts composing it. A Unit also possesses a role and statistical attributes, all derived from its Artifacts.

Entry Tickets Other NFTs, the Entry Tickets (ETs), are mainly used to participate in competitive challenges. Depending on the level and nature of the rewards, a challenge can require different levels or amounts of ETs. The main way to collect ETs is to be ranked in the PvP and PvEvP challenges (top ladder). An alternative is to purchase ETs on the marketplace, where they are tradeable. There are four levels of ET rarity: Bronze, Silver, Gold, Diamond. Special events, challenges, or tournaments can require dedicated entry tickets. The maximum amount of ETs is not capped.

3.2.3 Subscriptions

Starter pack The Starter Pack is a unique offer, once in a user’s account lifetime, with the highest ROI value across the game. For an exceptionally low price, typically \$3 to \$7, it gives the gamer a leg up by providing them with Time Stones, some Time Fluid and XP Dust, but also game tokens. It first pops up after a tutorial phase lasting 5 to 20 minutes, then is presented again at regular intervals during the player’s first few sessions, and remains available in the shop if it has not been taken advantage of so far.

Battlepasses “Battlepasses” are time-limited subscriptions or memberships that allow players to unlock and earn dedicated rewards based on their progress (activities and results). A Battlepass is a strong monetization and retention feature, which requires players to regularly reconnect to complete required activities. They must pay upfront in tokens or fiat currency, usually for a period of 30 days. Battlepasses are not tradeable or shareable. Players have to complete precise objectives to earn rewards. Rewards are distributed on the course of different milestones. Completing all milestones of a battlepass is a long-term and difficult challenge.

There are three types of Battlepass: Free, Premium, and Legendary. In term of economics, the Battlepass approach enables the precise regulation and control of rare

and Web3 rewards with respect to the engagement time (sessions and session durations). Buying a Battlepass generates and injects regular revenues into the different reward pools and toward the company's ROI.

- **Free Battlepass:** Any user has access to the Free Battlepass and its rewards, which are limited but still attractive from a non-paying user's perspective (see next paragraph). Here the main objective is to "onboard" in the blockchain sense traditional free-to-play (F2P) and play-for-fun (P4F) players so they can start interacting with Web3 features such as basic minting, and are able to earn a limited amount of tokens. The different types of Free Battlepass rewards are: (i) Bronze Entry Tickets, with which players can participate in low-reward competitive challenges (PvP, PvEvP); (ii) a few Time Stones, with which players can mint a few Units for free; (iii) a few rare Artifacts to start a collection of rare items; and (iv) temporary XP Dust boosters, which give players increased XP over a relatively short period.
- **Premium Battlepass:** This is a paying Battlepass designed for competitive P4F players and moderate Web3 "grinders" (P2E players). The main goal of the Premium Battlepass is to encourage participation in competitive challenges by regularly allowing them to earn top-tier PvP and PvEvP Tickets (Silver, Gold), and by removing Entry Ticket requirements for lower-tier PvP and PvEvP challenges (Bronze). The Premium Battlepass provides three Epic Artifacts upfront (pre-release channel). Similar to its Free version, this Battlepass also distributes various rewards to the player with better perks, such as instant Energy Point boosters.
- **Legendary Battlepass:** This VIP-grade Battlepass is made for extremely competitive Web3 grinders. The Legendary Battlepass confers all of the Premium privileges, plus dedicated ones. Its main goal is to facilitate participation in highly rewarding challenges through VIP fight Tickets (Gold, Diamond) and waiving Entry Tickets for low-to-medium challenges (Silver). It provides Legendary Artifacts upfront and all sorts of rewards as above.

3.3 Game Economy

3.3.1 General principles

The economy of *Time Breachers* was designed around three core principles: multi-level experience, equal opportunity, and incentivized involvement.

Multi-level experience The game proposes different levels of gaming experience on which players choose their financial involvement. Players are tagged into four categories: casual, midcore, hardcore, and Web3 natives. Each category follows a custom core loop, where more financial involvement is associated with longer playing sessions and improved perspective to win significant in-game rewards.

Equal opportunity All users, including F2P players, are offered a first-rate gaming experience. They can reach the same levels and have the same odds of winning with time or money, or both. They can all collect in-game items and level up. Even non-paying users

may earn tokens and common NFTs in the long run. Time-rich and money-rich players can experience the same quality of gaming, just not at the same pace.

Incentivized involvement Gamers with an initially limited engagement are entrained and incentivized toward different consumption habits and cycles, including collecting in-game resources, crafting Artifacts and Units, using them in action, and burning them to obtain more powerful crafting resources. This training facilitates the transition to the paid area of the game.

3.3.2 Long-term economic sustainability

The game's long-term sustainability is made possible by the following three features: a hybrid business model, a mitigated inflation, and a proxy currency.

Hybrid business model A hybrid business model was adopted in which financially involved users purchase entry tickets, battlepasses and boosters to access the most rewarding parts of the game. To ensure sustainability, each activity that involves an opportunity of in-game asset rewards (Time Stones, Artifacts, even Uchronians) and most PVP tournaments all require a specific entry ticket. Optional and low-price subscription formulas create smooth and meaningful paywalls while incentivizing F2P users to embark on the paying side of the game.

Mitigated inflation NFT inflation is controlled by incentivizing players to mint, play, and burn their NFTs within the core game loop. Special attention is paid to the creation of sufficient and attractive NFT- and resource-sinks to counterbalance the sources.

Single-token economics The game relies on a single-token model (see below). The Time Stone ERC-20 token is both the main utility token of the ecosystem and the governance token. Its value is strengthened by a limited supply and its use for important in-game and on-chain actions such as summoning Artifacts and Uchronians, or purchasing PVP tickets. They are also used to reward non-paying and paying players and help them progress through the game, giving them a say in the governance.

3.4 Game Tokenomics

Following the discussion of Section 2.5 about the comparative merits of single-token versus dual-token economies, we consider that the single-token model is the appropriate choice to support the creation of a sustainable and financially viable economy in this game. The market value of Time Stones will be strengthened by the concentration of several uses, the main ones being its dual role as a share token and an in-game currency. This advantage is outlined in a formal model described in Appendix A.

3.4.1 The single token as a share token

Time stones (\$TSTs) will serve as the protocol's share token. They are characterized by a limited supply and an emission schedule algorithmically enforced by the genesis smart

contract. \$TSTs provide token holders with claims to shares of the revenue captured by the game via different means:

- sales of starter packs, battlepasses, and entry tickets
- 5% to 100% fees on some marketplace transactions or crafting actions.

Initially, all revenues go to the game's Treasury (Section 3.4.3). Usually, in a later stage, when \$TST holdings become sufficiently decentralized among retail investors and players, the share token of a protocol finds a second use as a governance token, giving users a voting power in the DAO. In the present game, however, governance privileges are acquired via the Governance Score, not directly by the share token. At any rate, governing members of the organization will be able to decide on changes or additions to the game contents, and to allocations from the Treasury, such as rewards from the staking contract or grants to grow the game and promote the onboarding of new players.

3.4.2 The single token as a currency

Second, the Time Stone will also serve as an in-game currency on the marketplace, where it will have legal tender. To control inflationary pressure, the distribution of in-game incentives in the form of \$TSTs will be limited to those players with the highest PVP values in the leaderboard. Additional game incentives will be provided by the distribution of valuable and mandatory resources within the core loop. Moreover, by not making the \$TST a mandatory input to engage in the game we also ensure that low-wealth players are not locked out by a rising token price. Additionally, an appreciating \$TST will mean more purchasing power to buy essential game items.

3.4.3 Token flow chart

The circulation of the unique game token is summarized by a schematic flow chart (Fig. 15). All the \$TST emission originates from the genesis supply, which gradually mints the tokens according to pre-agreed vesting arrangements. The supply will be partly distributed to investors and team members, partly given away to players as game incentives and to liquidity providers as a second source of yield. A fraction is kept in the DAO's Treasury. \$TSTs emitted for in-game rewards are stored in a reward fund before being injected into the game. This allows for the effective emission to be tuned by the development team, while preventing excess distribution not originally planned in the genesis contract. When the game's governance becomes fully decentralized, this responsibility will be eventually taken over by the DAO.

Fees and sales collected on the marketplace, whose volume will progressively increase with the growth of the user base, are deposited in the DAO's Treasury. The game will also implement an in-game decentralized exchange (DEX) in which Time Stones will be tradeable against a base asset. \$TST holders will be incentivized to stake their tokens in the DEX. LP tokens deposited in the staking contract will earn a high APR interest rate in the form of \$TSTs. Most of the additional yield will be initially provided by the genesis contract according to a programmatic emission schedule. As the user base expands, the extra yield will be increasingly provided by organic revenues accumulated in the DAO's Treasury.

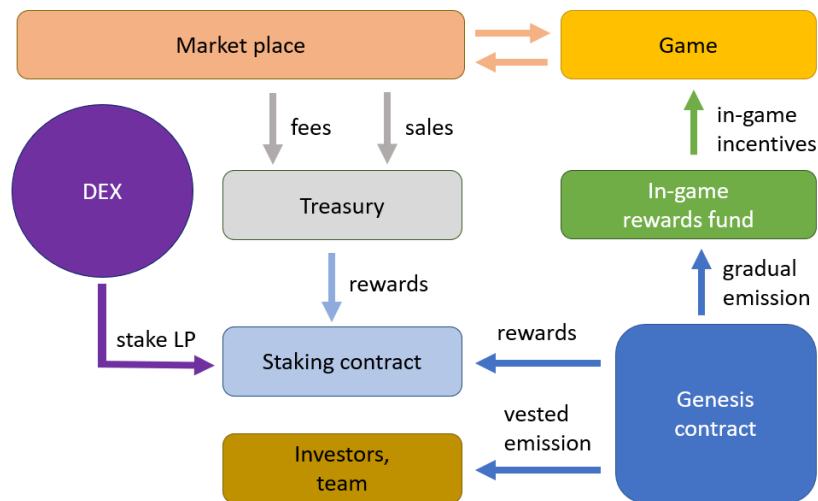


Figure 15: Starchain Gazer's "Time Stone" (\$TST) token flowchart.

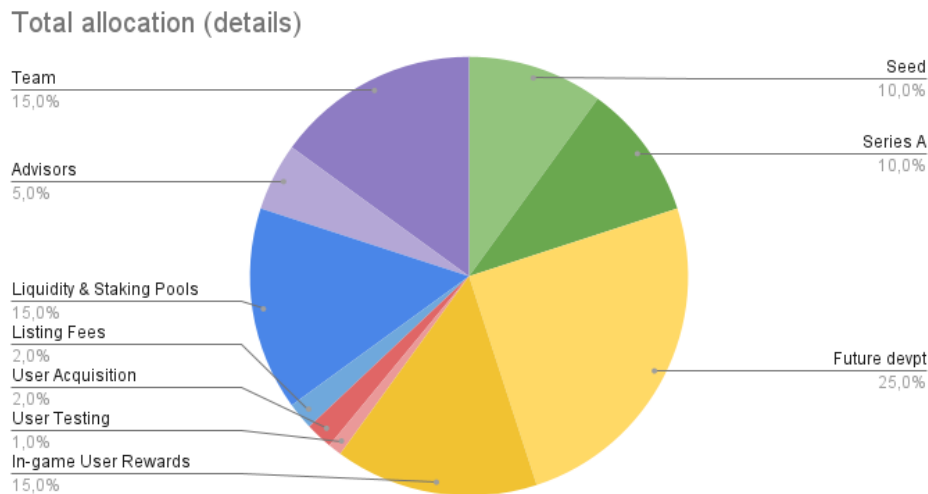


Figure 16: Starchain Gazer's token allocation.

3.4.4 \$TST distribution

To bootstrap game adoption, the tokenomics will be strongly user-oriented: 18% of the supply will be dedicated to gaming incentives and player acquisition. Overall, the token distribution is composed of five main parts (Fig. 16):

- 20% for Starchain Gazer = team (15%) + advisors (5%)
- 17% for DeFi = liquidity & staking pool (15%) + CEX listing fees (2%)
- 18% for users = in-game rewards (15%) + user acquisition (2%) + testing (1%)
- 20% for investors = seed (10%) + series A (10%)
- 25% for future development & marketing

A time-dependent emission schedule also informs on how the \$TSTs are gradually emitted (Fig. 17). The supply distributed to investors is vested 48 months, while the team and advisors' supply is locked 6 months then vested 42 months (with a 5% cliff). Marketing and in-game rewards, comprising future development, rewards, user testing and acquisition, are vested 60 months.

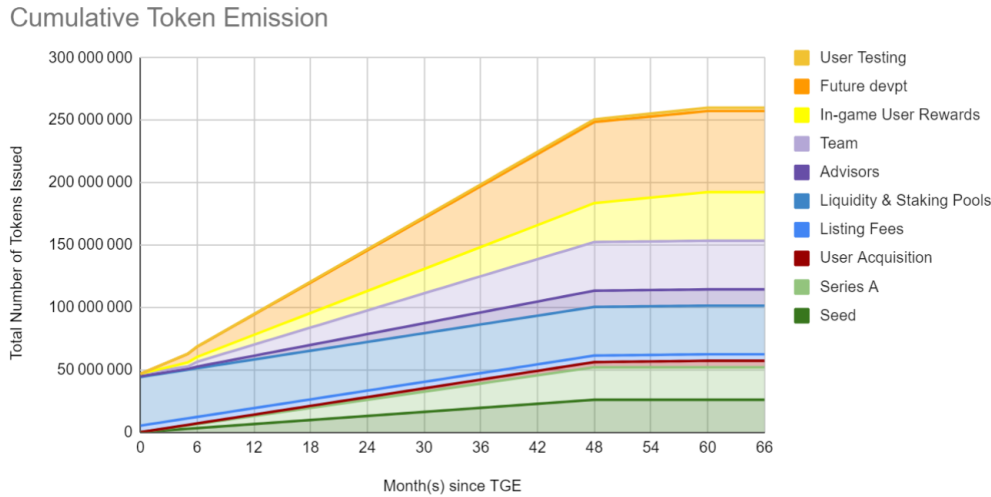


Figure 17: Starchain Gazer's cumulative token emission.

3.5 Projected revenue

A player's "lifetime value" (LTV) [9] is a key performance indicator. It shows how much revenue incoming users will bring during their whole period of involvement with the game. Given the strong heterogeneity among potential players, LTVs are calculated separately for each player category: casual players who opt for the F2P version, midcore players, hardcore players, and Web3 natives who buy battlepasses and items on the marketplace to benefit from advanced gaming experience.

First, we explain the assumptions about the player retention rate and its consequence on the daily number of players. Then, we multiply this number by the estimated daily fees and costs in each category. Finally, we add everything up to produce the estimated total revenue per year. (All numbers presented here are plausible hypotheses elaborated from an analysis of market user data; they are bound to change as this study evolves to reflect the maturation of the game design, itself based on continual feedback from the market.)

3.5.1 Model of player retention schedule

Based on standard references from the gaming industry [14, 6], we use the following retention rates per category of players (see also [4, 8]):

Retention/category after:	1 day	1 week	1 month	3 months
Casual players	45%	20%	10%	3%
Midcore players	55%	28%	10%	3%
Hardcore players	35%	15%	10%	1%
Natives	60%	40%	25%	2%

This table reads as follows: 55% of an initial population of midcore gamers continue to play the second day; this fraction goes down to 28% the second week, 10% after one month, and reaches 3% at the quarter mark, beyond which it remains constant. Given the continuous development and improvement of the gameplay, we assume that players engage in the game for a total duration of "playability" of up to five years, after which the retention rate finally drops to 0%. In summary, if we denote the midcore retention curve by $R_{\text{mid}}(d)$, where d is the index of the day, it is instantiated by the following table:

day d	1	2	8	31	91	366	...	1825*	1826
$R_{\text{mid}}(d)$	1	0.55	0.28	0.10	0.03	0.03	...	0.03	0

(* ignoring leap days). The complete daily schedule is an interpolation of these values using a geometric recursive law until Day 31, which produces an decreasing exponential form, then a linear approximation until Day 91 followed by a long plateau (Fig. 18).

Player retention schedule (projected)

(proportion of new players who joined at day D remaining after X days)

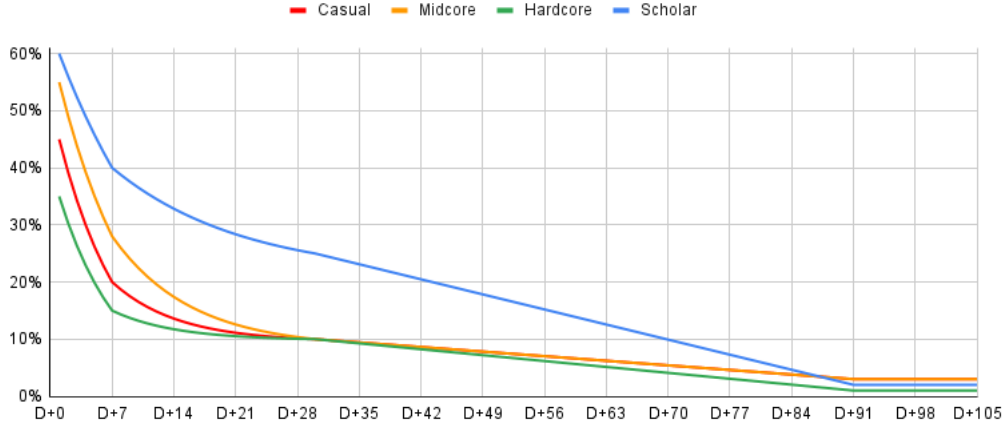


Figure 18: Projected players' retention schedule.

3.5.2 Projected evolution of the total user base

Assuming an effective marketing campaign, we plan to attract around 15,000 players at launch. Then, we assume that the number of new users increases by a variable influx $N(d)$ every day contributing to a gradual build-up of the total user base $UB(d)$ (the turnover fraction of “daily active users” (DAUs) among them, i.e. users who are actually playing that day, is not modelled here). We calculate UB using the retention rate curves defined above. Briefly put, it is the sum of the four per-category UB curves:

$$UB(d) = UB_{\text{cas}}(d) + UB_{\text{mid}}(d) + UB_{\text{hard}}(d) + UB_{\text{nat}}(d)$$

where each term is obtained by superimposing the corresponding retention curve R upon itself with a one-day shift every day and an amplitude equal to the number of new players N arriving that day:

$$UB_{\text{cas}}(d) = \sum_{d'=1}^d N_{\text{cas}}(d') R_{\text{cas}}(d - d' + 1)$$

and same with the other three categories. If the numbers of new players in each category N_{cat} were constant, i.e. did not depend on the day d , the resulting curve UB would show a regular rise gradually slowing down and converging toward a constant plateau. In reality, one should take into account two major variations:

- The total number of daily new players $N(d)$ is expected to fluctuate wildly, as user acquisition growth is impacted by both micro and macro events: the former

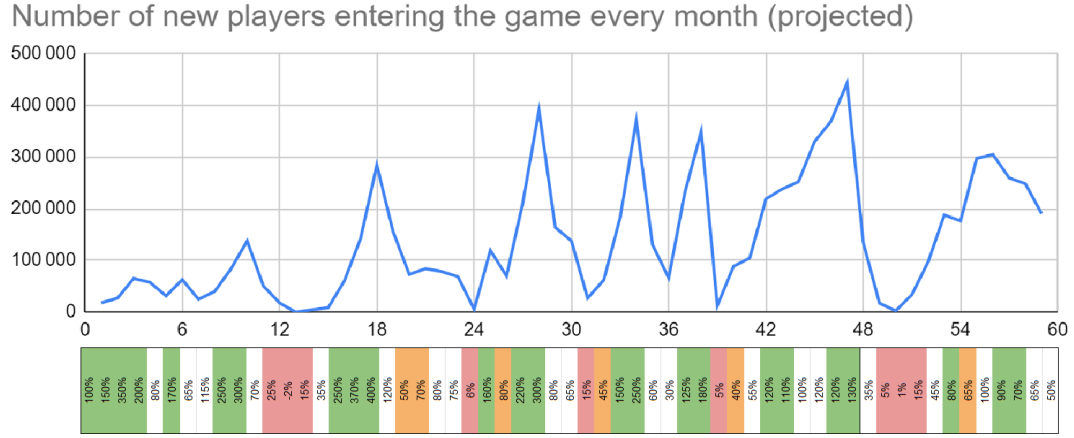


Figure 19: **Projected evolution of the number of new players.**

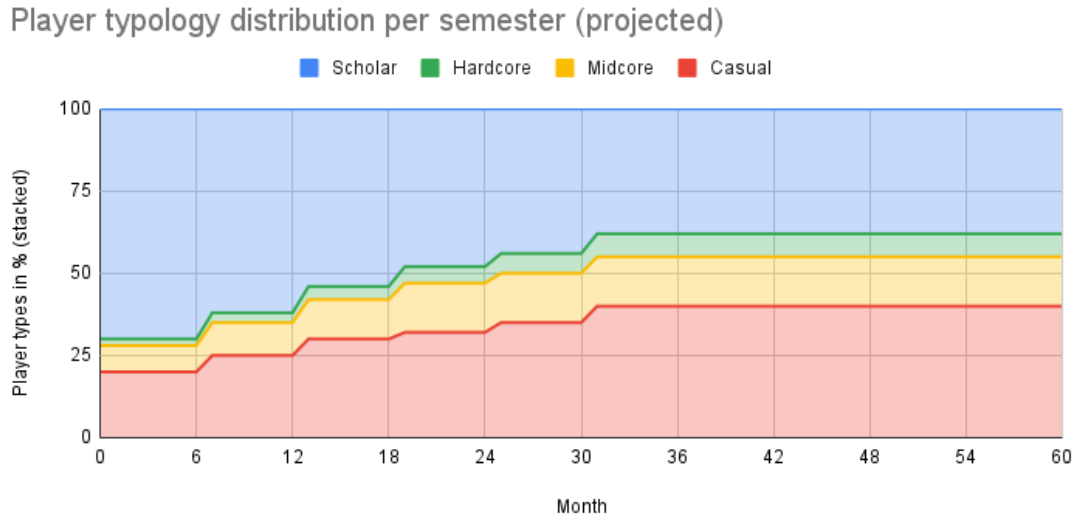


Figure 20: **Projected players' typology distribution.**

represent efforts to create value for gamers within the gameplay itself, e.g. through new releases; the latter may include various outcomes such as launches from the competition, booms and crashes of the crypto and financial markets, or new DeFi regulations. A possible projection is displayed here by month (Fig. 19).

- Moreover, *within that total* the respective shares of casual, midcore, hardcore and Web3 native players N_{cat} will also vary over the lifetime of the game, most likely displaying an overwhelming majority of Web3 natives in the beginning, then a gradual increase in casual players until rough equality between the two camps, while the midcore and hardcore tiers tend to shrink; in short:

$$N_{\text{cas}}(d) = \lambda_{\text{cas}}(d)N(d), \dots, N_{\text{nat}}(d) = \lambda_{\text{nat}}(d)N(d)$$

where the four coefficients obey $\lambda_{\text{cas}} + \lambda_{\text{mid}} + \lambda_{\text{hard}} + \lambda_{\text{nat}} = 1$, and are susceptible to evolve as in Fig. 20 (by semester).

Putting everything together, the stacked curves of Fig. 21 show the progression of the user base UB_{cat} in each category and their sum UB (top curve) over the first five years.

Average daily user base per month (projected)

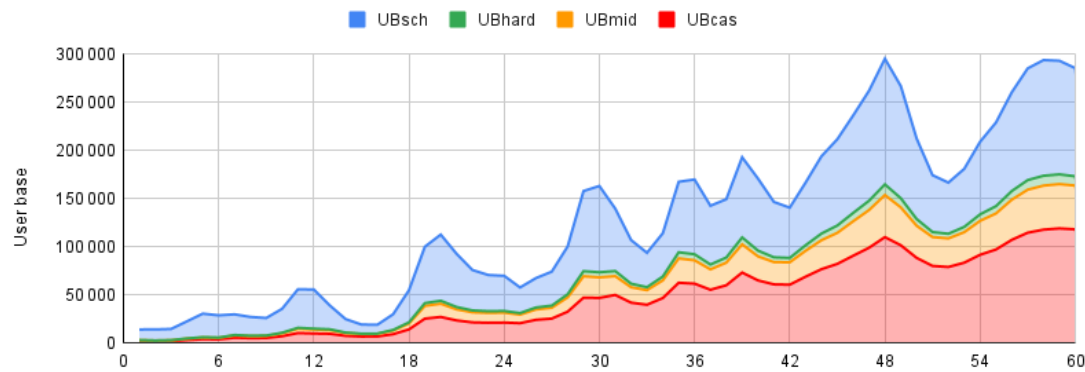


Figure 21: Projected average daily user base.

3.5.3 Estimated fees and sales per user

To obtain the global revenue, the last component we need is an evaluation of the average fees and sales likely to be cashed in from each player of a given category per day or per month. Then we can compute the weighted sum of these fees multiplied the above $UB_{<cat>}$ functions. These fees are themselves the product of three factors: a base price, a frequency of buying/selling per session, and an average number of playing sessions per day.

Base prices At every session, players go through various steps, some of which generate revenues for the game:

- Starter packs: these packs provide players with a set of basic items that facilitate their transition from F2P to P2E; we set their individual price to \$3. They are bought only once by an estimated 2% of the new players.
- Battlepasses: the prices of the Premium and Legendary battlepasses are set to \$5 and \$15, respectively.
- Time Stones: players have the possibility of buying \$TSTs directly from the marketplace, in addition to collecting them in the game; the initial exchange rate is estimated at \$1 apiece.
- Marketplace: the marketplace allows players to buy and sell Artifacts and Units from one another. Units can also be temporarily borrowed and loaned between players. The game applies a blanket 5% fee on all transactions.
- Other actions such as crafting Artifacts & Units, and upgrading Units & Motherships are paid in \$TSTs by the players, 50% of which go to the share holders.
- Various grades of “Entry Tickets” to PVP sessions are also available for sale (bronze, silver, gold, platinum, diamond, tournament) at prices ranging from \$0,30 to \$12.

In addition to these daily gamer activities, SCG will organize a special one-day sale of NFTs, first on the date of the official launch, then twice a year, with an increase in quantity of 25% per year:

NFT Rarity	Quant.	Price \$ETH	Total \$ETH	Year 1 \$USD*	+25%	Year 5 \$USD*
Common	15,000	0.01	150	\$240k	...	\$586k
Rare	5,000	0.05	250	\$400k	...	\$976k
Epic	2,500	0.1	250	\$400k	...	\$976k
Legendary	500	0.5	250	\$400k	...	\$976k
TOTAL	23,000	—	900	2×\$720k	...	2×\$1757k

(*) assuming an average exchange rate of \$ETH 1 $\approx$ \$USD 1600.

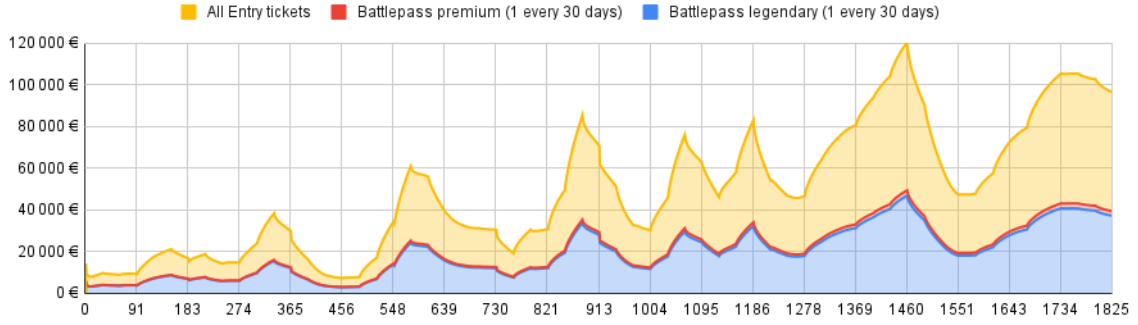
Actions per day The estimated frequencies, or probabilities, of a player to buy/sell an entry ticket or battlepass, and the estimated average quantities, or fractions, of other in-game items bought/sold or borrowed/loaned during one session, are shown in this table:

Action frequency/player/day	Casual	Midcore	Hardcore	Native
Starter pack (once/5 years)	2%/1825	2%/1825	2%/1825	2%/1825
Premium battlepass (1/month)	—	9%/30	40%/30	5%/30
Legendary battlepass (1/month)	—	1%/30	30%/30	65%/30
Time Stone bought	0.1%	0.4%	2%	0.7%
Artifact bought/sold	—	1.25%	4%	10%
Unit bought/sold	—	2.5%	20%	20%
Unit borrowed/loaned	—	—	4%	20%
Artifact crafted	7.5%	25%	40%	40%
Unit potential upgraded	0.75%	1.25%	8%	8%
Unit crafted	0.75%	2.5%	6%	6%
Mothership enhanced	0.75%	1.25%	4%	6%
Entry tickets bought	1.5%	12.5%	40%	40%

Resulting fees and sales Combining the above base prices and action frequencies, we obtain the following revenue estimates in the game core loop per player and per day, denoted by $R(d)$ (here, excluding the starter packs and NFT sales):

Revenue/player/day	fees	R_{cas}	R_{mid}	R_{hard}	R_{nat}
Premium battlepass	\$5.00	—	¢1.50	¢6.67	¢0.83
Legendary battlepass	\$15.00	—	¢0.50	¢15.00	¢32.50
Time Stone bought	5% $\times$ \$1.00	¢0.005	¢0.02	¢0.10	¢0.035
Artifact bought/sold	5% $\times$ \$0.45	—	¢0.03	¢0.09	¢0.225
Unit bought/sold	5% $\times$ \$1.05	—	¢0.13	¢1.05	¢1.05
Unit borrowed/loaned	5% $\times$ \$0.90	—	—	¢0.18	¢0.90
Artifact crafted	50% $\times$ \$0.90	¢3.38	¢11.25	¢18.00	¢18.00
Unit potential up	50% $\times$ \$0.30	¢0.11	¢0.19	¢1.20	¢1.20
Unit crafted	50% $\times$ \$1.80	¢0.68	¢2.25	¢5.40	¢5.40
Mothership enhanced	50% $\times$ \$0.30	¢0.11	¢0.19	¢0.60	¢0.90
Entry tickets (avg)	50% $\times$ \$0.3,...,\$12	¢0.25	¢5.06	¢47.70	¢45.60
TOTAL		\$0.045	\$0.21	\$0.96	\$1.07

Revenue per day (projected): tickets & battlepasses



Revenue per day (projected): misc. fees, except NFT sales

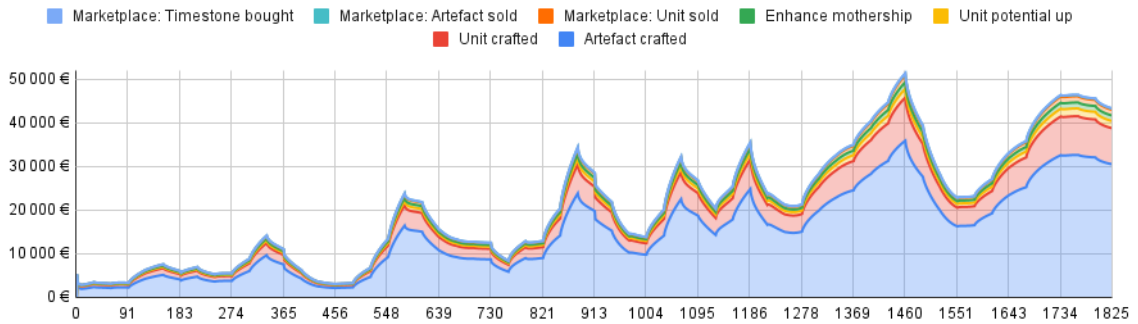


Figure 22: Projected revenue per day.

3.5.4 Game's global revenue

Finally, this last table is combined with the four UB curves (Section 3.5.2) into a weighted sum per day, to which starter packs are added via multiplication with the number of new players, plus exceptional NFT sales twice a year, to produce the global revenue per day (Fig. 22).

$$GR(d) = R_{cas}(d).UB_{cas}(d) + R_{mid}(d).UB_{mid}(d) + R_{hard}(d).UB_{hard}(d) + R_{nat}(d).UB_{nat}(d) + S.N(d) + (d \bmod 180 \neq 0).NFT \quad (1)$$

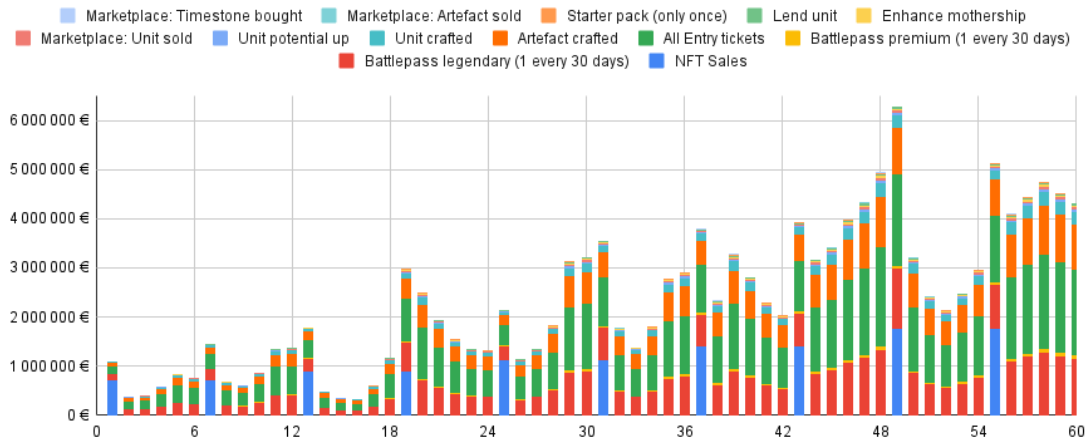
The same projection can be renormalized on a monthly basis, then integrated to produce the cumulative earnings over five years (Fig. 23). In this model, the final numbers are:

Year	Fees/year	NFTs/year	Revenue/year	Cumul. revenue
1	\$8,975,490	+ \$1,440,000	= \$10,415,490	\$10,415,490
2	\$14,573,096	+ \$1,800,000	= \$16,373,096	→ \$26,788,586
3	\$24,790,377	+ \$2,250,000	= \$27,040,377	→ \$53,828,963
4	\$37,527,487	+ \$2,812,500	= \$40,339,987	→ \$94,168,950
5	\$43,292,980	+ \$3,515,625	= \$46,808,605	→ \$140,977,555

3.6 Modelling the Demand for Time Stones

In accordance with the single-token economics, Time Stones play two roles at once: (a) on the one hand it is the *in-game currency*, which has legal tender in the marketplace and is used as means of payment for various game actions such as item crafting or upgrading;

Revenue per month (projected)



Cumulative revenue by month (projected)

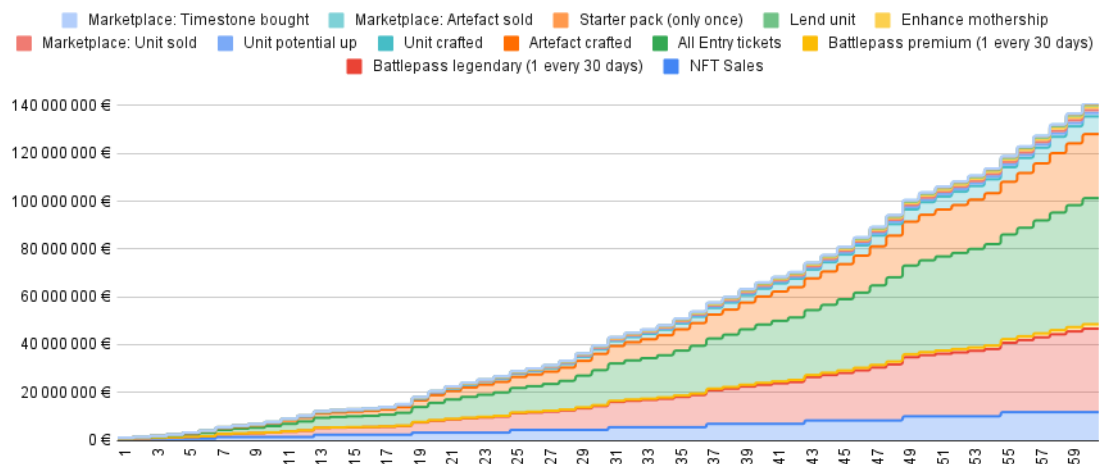


Figure 23: **Projected revenue and cumulative revenue by month.**

and (b) on the other hand it is also the *share token*, which accrues value from the game. As a consequence, the demand for Time Stones (\$TST) involves two components:

- Demand for transaction purposes: this first component arises from the absence of perfect synchronization between payments and earnings in \$TST. Players spend \$TST, for example, to craft Artifacts, Uchronians or buy NFTs in the marketplace, while on the contrary they gain \$TST by burning Uchronians or Artifacts. Therefore, holding \$TSTs in their wallet allows them to bridge the gap between \$TST inflow and outflow.
- Demand for staking purposes: this second component of demand is motivated by the opportunity offered by the game to deposit \$TSTs in the staking module and earn some return.

We quantitatively evaluate the two components of demand. Regarding the transaction component, we assume that players set aside the equivalent in \$TSTs of one day of spending. First, we give an estimation of how many actions (crafting, buying, and so on) and their cost in \$TSTs each category of players achieve every day. Then, we extrapolate daily aggregate spending by factoring in the projected population of players.

3.7 Modelling the Demand for Staking

Staking opportunities are introduced in the game to provide players with in-game rewards. In addition to \$TST tokens given out as in DeFi protocols, a gamification layer also grants stakers free in-game utility items such as PVP tickets. These advantages will change with each game season to boost the attractiveness of the staking module.

To estimate the demand for staking, we adapt the general model of Appendix A, in which the game gradually distributes its token to the market. Circulating supply at dates $t = 0, 1, \dots$ is denoted by s_t . Initial circulating supply is s_0 , then gradually converges to its maximum $\bar{s}$. Two types of income are distributed to stakeholders in the staking module:

- fees y_t collected from the game
- freshly minted supply dedicated to liquidity incentives, valued at current price: $p_t x_t$

The discounted cash flow model prices the protocol as the present value of future incomes discounted at rate r :

$$p_t = \sum_{z=1}^{\infty} \frac{1}{(1+r)^z} \frac{y_{t+z} + p_{t+z} x_{t+z}}{s_{t+z-1}}$$

Stakeholders receive protocol's revenue akin to a dividend y_{t+z}/s_{t+z-1} , but contrary to traditional firms, they also benefit from the continuous distribution of tokens.

The price earning ratio (PER) is a convenient measure of performance. It is derived by dividing the price p_t of the token by protocol's earnings per token y_t/s_t . A high PER means that investors are expecting high growth rates in the future; it reflects in the price but not yet in current earnings. The PER defined by $\pi_t = p_t/(y_t/s_t)$ thus reads:

$$\pi_t = \sum_{z=1}^{\infty} \frac{1}{(1+r)^z} \frac{y_{t+z}}{y_t} \frac{s_t}{s_{t+z-1}} \left(1 + \frac{x_{t+z}}{s_{t+z}} \pi_{t+z} \right)$$

It has two components: an earning component coming from the protocol's organic profitability and a token emission component.

On the long run, all tokens are distributed and the circulating supply reaches its maximum. With an emission x_{t+z} converging to zero and both s_t and s_{t+z-1} converging to $\bar{s}$, the PER becomes the discounted sum of income growth rates:

$$\pi_t = \sum_{z=1}^{\infty} \frac{1}{(1+r)^z} \frac{y_{t+z}}{y_t}$$

Assuming that the income y_t grows at a constant rate g , pricing is given by the classical Gordon growth formula:

$$\pi_t = \sum_{z=1}^{\infty} \left(\frac{1+g}{1+r} \right)^z = \frac{1+g}{r-g}$$

It means that the higher the protocol's growth rate g , the higher its PER. On the short and mid-terms, the token emission adds a second source of income, preventing closed-form solutions, but the PER, the return rate and the price can still be numerically simulated.

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Appendix A: Modelling a Single/Dual-Token Game Economy

This section presents a simplified model of gaming economy with one or two tokens: an in-game currency (IGC) and/or a share token (ST). We evaluate the capacity of each system to bring value to token holders on the premise that the game attracts a growing base of players. We begin by introducing the IGC for pedagogical reasons.

A.1. Game Economy with In-Game Currency

We set up a simplified economy composed of two assets: an IGC and an external asset which serves as numeraire (e.g. \$ETH or \$USDC). To access the game, players acquire d_t units of the IGC at price q_t . One period later, they get $(1+r)d_t$ where r is the reward from playing, then they sell at price q_{t+1} and exit the game. The demand for IGC is inversely related to its price:

$$d_t = \frac{d}{q_t}$$

where $d > 0$. The flow of new players n_t is growing with time at a constant rate g : $n_t = (1+g)n_{t-1}$. The number of active players at each date is then: $x_t = n_t - n_{t-1} = gn_{t-1}$ which is also growing at rate g .

The IGC price is determined by the demand-equal-supply equilibrium condition: $n_t d_t = n_{t-1} d_{t-1} (1+r)$ or

$$\frac{n_t d}{q_t} = \frac{n_{t-1} d (1+r)}{q_{t-1}}$$

On the left-hand side, the demand for IGC comes from new players entering the game, each buying d_t units. On the right-hand side, the supply comes from players exiting the game and selling their $(1+r)d_{t-1}$ units. Therefore, the equilibrium market price growth rate is:

$$\frac{q_t}{q_{t-1}} = \frac{1+g}{1+r}$$

It means that the higher the price increase rate, the higher the user base growth rate g , and the smaller the rewards rate r designed to incentivize the game. In this simple economy, a new IGC emitted at rate r is not inflationary if it matches the economy's growth rate: $r = g$.

In-game rewards play an essential role in stabilizing the IGC price. Without them, it would increase over time making the game unaffordable for many users. At the same time, an excessively high rewards rate would lead to inflationary pressure and a downward trending price.

An interesting property of this simple game example is that whatever the rewards rate r , the return from playing does not depend on it but only on the user base growth rate:

$$\frac{(1+r)q_{t+1}}{q_t} = 1+g$$

Here the player buys one unit of resource at price q_t and sells $1+r$ unit at price q_{t+1} one period later. The higher the rewards rate, the higher the IGC inflation rate, which cancels the higher rewards off.

A.2. Adding a Share Token

It is possible to augment this simple economy by adding an ST. There is a treasury controlled by a DAO, toward which fees y_t collected within the game are channelled. Buy and sell operations in the marketplace are taxed at rate τ :

$$y_t = \tau n_t q_t d_t = \tau n_t d$$

Assuming that the fees growth rate is equal to the game's growth rate g , the return from playing is reduced by the fee rate as follows:

$$(1+r)(1-\tau)\frac{q_{t+1}}{q_t} = (1+r)(1-\tau)\frac{1+g}{1+r} = (1-\tau)(1+g)$$

which means less demand for the IGC and less gaming activity.

Contrary to the IGC, which has an uncapped supply, the ST has a fixed supply (set to 1 here). Denoting by p_t the ST price and applying the discounted cash flow pricing model, we obtain:

$$p_t = \sum_{k=1}^{\infty} \frac{y_{t+k}}{(1+\rho)^k} = \sum_{k=1}^{\infty} \frac{y_t(1+g)^k}{(1+\rho)^k} = \frac{1+g}{\rho-g} \tau n_t d$$

where $\rho > g$ is the discount rate. The ST price growth rate equals the game's growth rate g . We obtain a well-behaved dual-token economy with: (a) an IGC whose price is stable insofar as the rewards rate tracks the game's growth rate; and (b) an ST whose price captures the user base growth rate through a tax on trades.

A.3. Single-Token Economy

A single-token economy is one in which the ST also serves as IGC. Its price is still determined by the discounted cash flow formula and the no-arbitrage condition. As in the two-token economy, if revenues generated by the protocol increase (decrease) beyond market expectations, higher (lower) return given to token-holders are arbitrated down (up) by a higher (lower) price.

The fundamental difference is the absence of a second token whose price balances supply and demand for in-game transactions. So, either demand for ST for in-game transactions is the short side of the market (and supply side is limited by the quantity demanded), or supply of ST is the short side (and demand is limited to the quantity supplied). Considering that the ST price grows at rate g and the IGC price at rate (about) $g - r$ in the two-token economy, the most likely outcome is that demand for in-game transactions is driven down by high price and supply is constrained by demand. This is not to say that ST holders cannot exit the game by selling their tokens but that they prefer to stake them to earn the staking return. The financial motive of holding ST dominates the transaction motive.

Fees are still a tax on in-game transactions:

$$y_t = \tau n_t p_t d_t = \tau n_t d$$

and fees growth rate is still equal to the game's growth rate g . We get the same cash-flow-based pricing formula:

$$p_t = \frac{1+g}{\rho-g} \tau n_t d$$

with $\rho > g$ the discount rate. The price growth rate is given by the players' growth rate g . The return from playing is now:

$$(1+r) \frac{p_{t+1}}{p_t} = (1+r)(1+g)$$

Game playing gives a double dividend: players get the reward r and benefit from price appreciation g . At the same time, demand for in-game transactions is shrinking with price increase:

$$d_t = \frac{d}{p_t}$$

With a price increase rate equal to g , the demand decrease rate is (about) $-g$. This does not affect the volume of fees though, as the decreasing demand at rate $-g$ is compensated by an increasing price at rate g .

This table summarizes the differences between the two types of economy:

	Two tokens	One token
Players increase rate	g	g
ST price increase rate	g	g
IGC price increase rate	$g - r$	$g - r$
Return to play	g	$g + r$
Transaction demand per player	$r - g$	$-g$

To recapitulate, a single token economy has only one ST, whose price increases with the number of players. This drives out demand for in-game transactions on the intensive margin (the average transactions per player), but not on the extensive margin (players are still willing to buy some ST to play or to earn the staking return rate). When those two effects are combined, the game's profitability is not impaired as fees are collected on a decreasing transaction volume but an increasing transaction price. This also makes the game more financially attractive for players as they buy a token which appreciates over time.

Appendix B: Tokenomics Compared

B.1. Gaming Tokenomics

	Axie Infinity	Illuvium
Upfront cost to E2P	3 Axie NFT	—
Share to gaming incentives	20%	10%
Share to staking incentives	29%	6%
Share to LP incentives	0%	24%
Gaming incentives time lock	0 to 54m	—
Staking incentives time lock	6 to 65m	—
LP incentives time lock	—	0 to 12m
Utility of primary token	<i>staking, governance, breeding</i>	—
Secondary token (2T)	SLP	—
Utility of 2T	<i>breeding</i>	—
Is ST marketable	yes	—
Pre-launch public sale	IDO 11%	Bal. LBP*

Table 1: Global characteristics. (*Balancer Liquidity Bootstrapping Pool)

	Axie Infinity	Illuvium	Star Atlas	Aurory
Private Sale	4%	—	22,5%	—
Public Pre-Sale	—	5%	1,5%	7%
Public Sale	11%	15%	2%	18%
Yield/Staking incentives	29%	30%	—	25%
Gaming Incentives	20%	10%	40%	same as yield/stk
Team & Advisor share	21% & 7%	15%	30%	15%
Ecosystem Fund	8%	—	—	7%
Treasury	—	15%	—	24%
Launchpad/Liquidity	—	10%	4%	4%

Table 2: Allocation comparison

	Axie Infinity	Illuvium	Star Atlas	Aurory
Private Sale	104	—	104	—
Public Pre-Sale	—	52	104	0
Public Sale	0	52	0	104
Yield/Staking incentives	282	156(+52)	—	312
Gaming Incentives	234	DAO*	416	same as yield/stk
Team & Advisor share	234 & 182	156	104	208
Ecosystem Fund	208	—	—	312
Treasury	—	DAO*	—	156
Launchpad/Liquidity	—	Bal. LBP	0	0

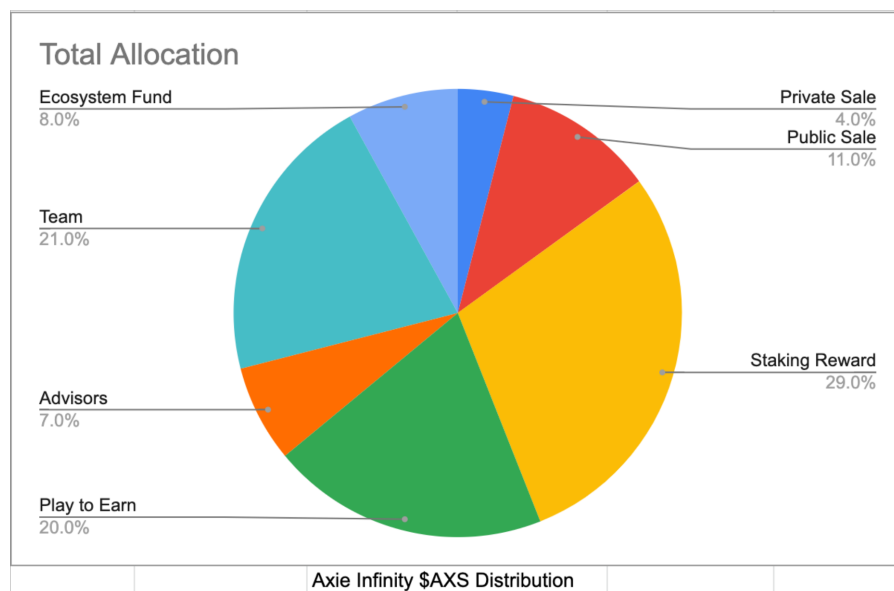
Table 3: Vesting time comparison in weeks. (*Decided by the DAO)

B.2. Axie Infinity

Governance token: \$AXS (total supply: 270,000,000). TGE: 03/11/2020.

Group	Allocation (%)	Allocation (\$AXS)	TGE Unlock	Vesting* Period (wks)
Private Token Sale	4%	10,800,000	20%	104
Public Token Sale	11%	29,700,000	100%	0
Advisors	7%	18,900,000	~23,8%	182
Staking Rewards	29%	78,300,000	0%	282 (start Q1-2021)
Gaming Rewards (P&E)	20%	54,000,000	0%	234 (start Q1-2021)
Team	21%	56,700,000	~19%	234
Ecosystem Fund	8%	21,600,000	~38,5%	208

Table 4: Axie Infinity allocation. (*Vesting: Process by which assets are locked and released slowly over a certain period of time.)

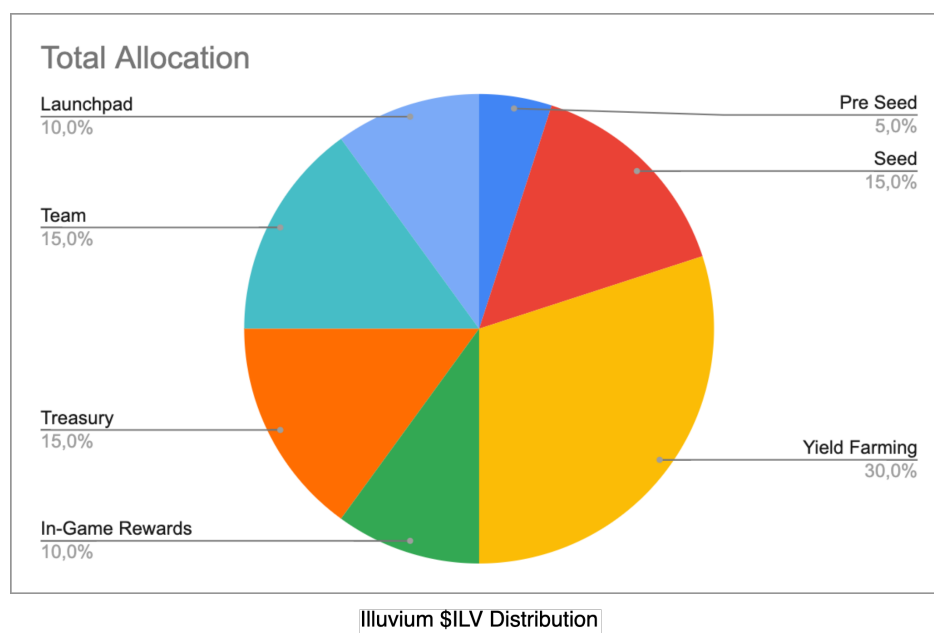


B.3. Illuvium

Ecosystem token: \$ILV (total supply: 10,000,000). TGE: 03/03/2021.

Group	Allocation (%)	Allocation (\$ILV)	TGE Unlock	Vesting Period (wks)
Pre-Seed	5%	500,000	0%	52*
Seed	15%	1,500,000	0%	52*
Yield Farming	30%	3,000,000	0%	156 (+52 after claim)
Gaming Rewards (P&E)	10%	1,000,000	?%	DAO decid.
Team	15%	1,500,000	0%	156*
Treasury	15%	1,500,000	?%	DAO decid.
Launchpad	10%	1,000,000	100%	Bal. LBP

Table 5: Illuvium allocation. (*start 03/2022)

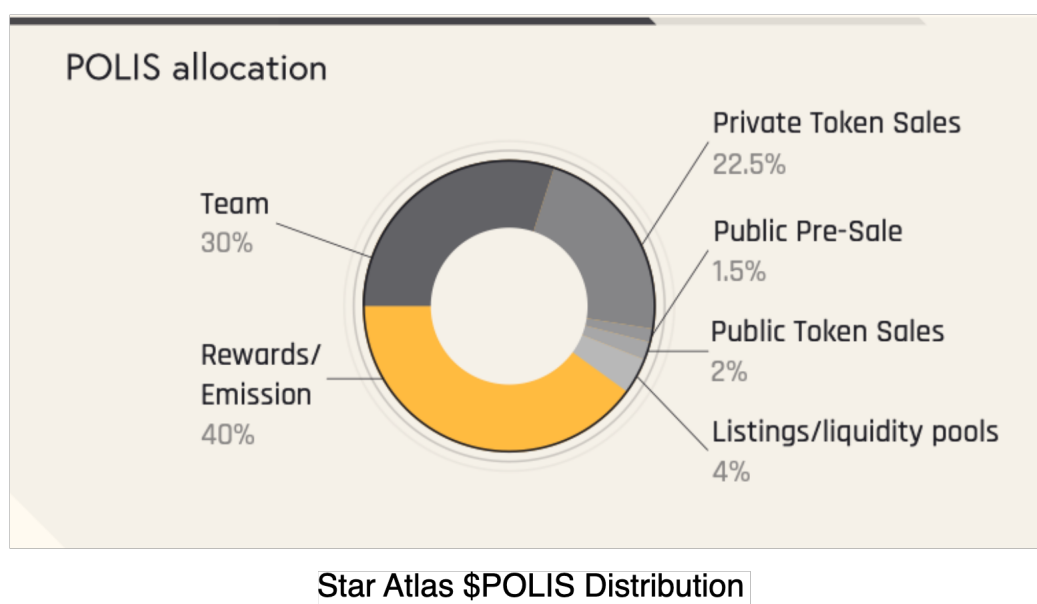


B.4. Star Atlas (Governance token)

Governance token: \$POLIS (total supply: 360,000,000). TGE: 26/08/2021.

Group	Allocation (%)	Allocation (\$POLIS)	TGE Unlock*	Vesting Period (wks)
Private Token Sales	22.5%	81,000,000	5%	104
Public Pre-Sales	1.5%	5,400,000	5%	104
Public Token Sales	2%	7,200,000	100%	0
Listings/Liquidity Pools	4%	14,400,000	100%	0
Rewards/Emission	40%	144,000,000	0%	416
Team*	30%	108,000,000	100%	104

Table 6: Star Atlas \$POLIS allocation. (*Unlock exclusively for staking in the DAO, 2-year linear vesting.)

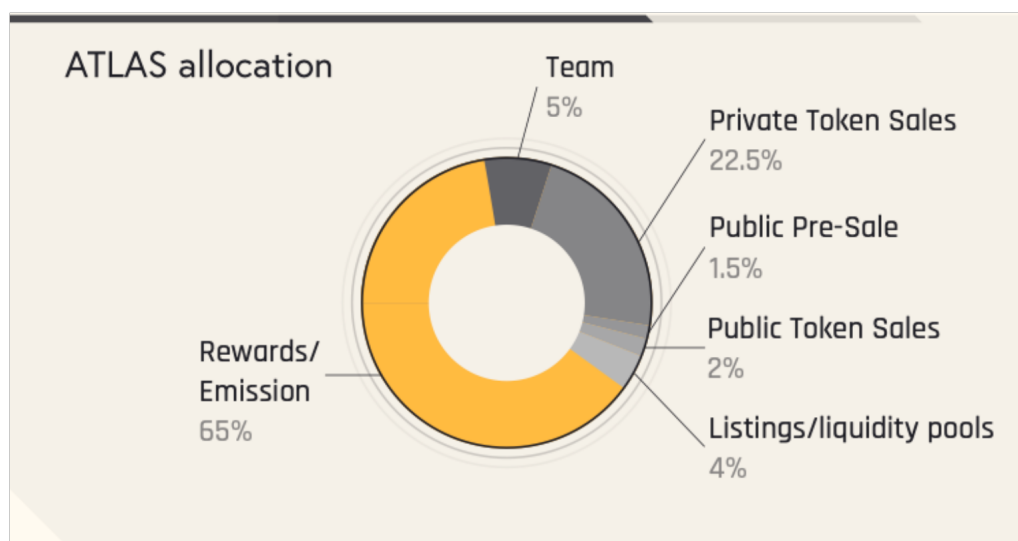


B.5. Star Atlas (Game token)

Game token: \$ATLAS (total supply: 36,000,000,000). TGE: 26/08/2021.

Group	Allocation (%)	Allocation (\$ATLAS)	TGE Unlock*	Vesting Period
Private Token Sales	22.5%	8,100,000,000	5%	104w
Public Pre-Sales	1.5%	540,000,000	5%	104w
Public Token Sales	2%	720,000,000	100%	0
Listings/Liquidity Pools	4%	1,440,000,000	100%	0
Rewards/Emission	65%	23,400,000,000	0%	416w
Team*	5%	1,800,000,000	100%	104w

Table 7: Star Atlas \$ATLAS allocation. (*Unlock exclusively for staking in the DAO, 2-year linear vesting.)



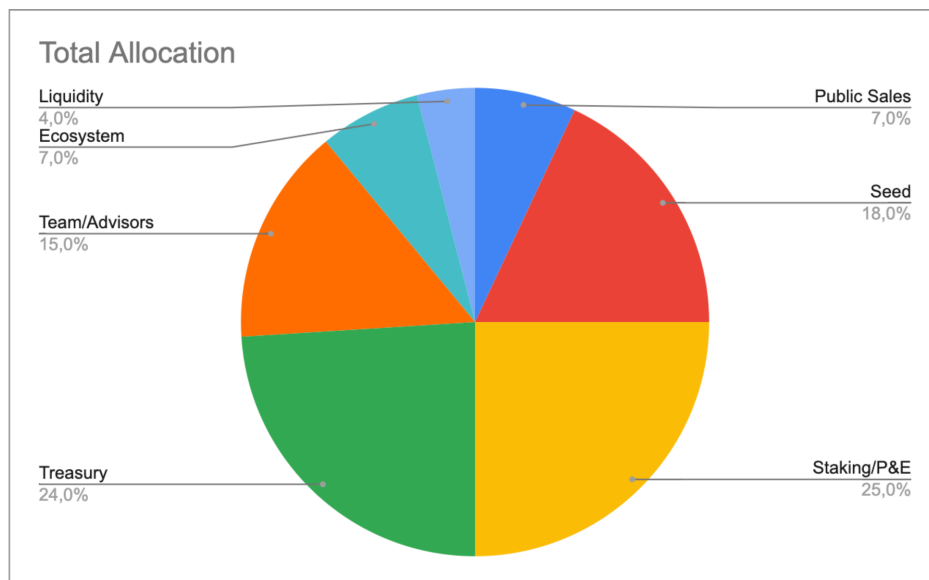
Star Atlas \$ATLAS Distribution

B.6. Aurory

Ecosystem token: \$AURY (total supply: 100,000,000). TGE: 26/10/2021.

Group	Allocation (%)	Allocation (\$AURY)	TGE Unlock	Cliff*	Vesting Period
Pre-Seed (Public Sales)	7%	7,000,000	100%	—	—
Seed	18%	18,000,000	0%	8M	2Y, Linear
Liquidity/MM	4%	4,000,000	100%	—	—
Yield Farming, Gaming	25%	25,000,000	0%	1M	6Y, Linear
Rewards (P&E)					
Team, Advisors	15%	15,000,000	0%	10M	4Y, Linear
Treasury	24%	24,000,000	0%	10M	3Y, Linear
Ecosystem	7%	7,000,000	0%	1M	6Y, Linear

Table 8: Aurory \$AURY allocation. (*Cliff: Period of time that must pass before the release of the tokens starts.)



Aurory \$AURY Distribution

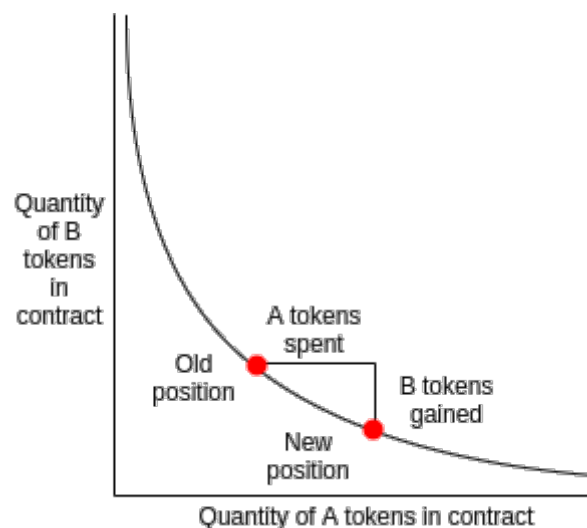
Appendix DS: SushiSwap

DS.1. Decentralized Exchanges

SushiSwap is originally a Decentralized Exchange (DEX) and Automated Market Maker (AMM) forked from the Uniswap V2 protocol. Like Uniswap, it manages a number of liquidity pools based on *pairs of currencies* (354 as of today). Unlike other exchange platforms, however, which guarantee and manage conversion pools centrally such as Binance or Kraken, a DEX incentivizes its community of users to collectively build the liquidities sought by clients via fees and other rewards in a way similar to lending. In general, the two main DEX rules are as follows:

- Creditors, also called “liquidity providers”, must deposit equivalent amounts of paired currencies forming a pool in terms of their value. In a pool \$A/\$B where the exchange rate of \$A vs. \$B is p , providers depositing x \$A must deposit $y = x/p$ \$B at the same time. For example, if $p = 2$, providers must deposit twice as many \$B’s as \$A’s:
 $y = 2x$.
- Moreover, token values are adjusted after every swap in such a way that the product of the total quantity of \$A contained in the pool multiplied by the total quantity of \$B remains constant: $x \cdot y = k$.

Therefore, if a client wants to convert \$A’s into \$B’s, the price of \$A, resp. \$B, will increase, resp. decrease, after the transaction is completed.



In exchange for depositing a pair of \$A/\$B quantities, liquidity providers receive pool-specific “A-B SLP Tokens”, where SLP stands for “Sushi Liquidity Provider”.

DS.2. Specificities

When it started in 2020, SushiSwap's rather cavalier way of bootstrapping their protocol made headlines, as it consisted of encouraging liquidity providers to stake *Uniswap LP tokens* into SushiSwap's own pools by rewarding them with \$SUSHI tokens. The issuance rate of \$SUSHI was set initially very high, up to 1000 tokens per block, with a schedule planned to go down to 100 within a year. Given the high market valuation of \$SUSHI at the time, the interest rate of many of these pools exceeded 2,500% APY per year. The goal was to transform, or "migrate" Uniswap's liquidities to SushiSwap—an aggressive strategy which was dubbed "Vampire Attack".

Today, SushiSwap rewards liquidity providers with fees collected from swaps, and with \$SUSHI tokens as follows:

- Each conversion transaction incurs a 0.3% fee: it is composed of 0.25% added to the pool and 0.05% transferred to another contract, the "Sushi Bar" (see below).
- The LP tokens can be staked for additional revenue paid out in \$SUSHIs, a process called *yield farming*.
- Furthermore, the earned \$SUSHI tokens can themselves be staked in the Sushi Bar in exchange for \$xSUSHI tokens.

DS.3. The Sushi Bar Contract

The Sushi Bar is a special pool where \$SUSHI holders can stake their tokens and receive in exchange a certain amount of \$xSUSHI tokens, a type of "LP" or "interest-bearing" token representing a proof of the staker's deposit similar to lending platforms (e.g., Aave's aTokens). \$xSUSHI accrues value thanks to the protocol fees mentioned above, as 0.05% of every swap is added to the Sushi Bar. At least once a day, a reward contract is called and all \$xSUSHIs are sold for \$SUSHIs. Staked \$SUSHIs are shared among the shareholders of the Sushi Bar pool in proportion to their deposits. By design, one \$xSUSHI should always have a greater value than one \$SUSHI.

To illustrate these rules, we assume that the staking pool contains a total of 1200 \$SUSHIs brought by three stakers Rose, Sam, and Tom. For that, they received a total of 300 \$xSUSHIs distributed as follows:

Stakers	\$SUSHIs	Share	\$xSUSHIs
Rose	400	$1/3 = 33.3\%$	100
Sam	200	$1/6 = 16.7\%$	50
Tom	600	$1/2 = 50.0\%$	150
TOTAL	1200	100%	300

At this point in time, the exchange ratio from \$SUSHIs to \$xSUSHIs is $1200/300 = 4$ for 1. Later, suppose that the Sushi Bar receives an extra 400 \$SUSHIs coming from fees collected on swapping transactions. These gains are distributed among the current three stakeholders in proportion to their shares, which do not change:

Stakers	\$SUSHIs	Share	\$xSUSHIs
Rose	$400 + 400/3 = 533$	33.3%	100
Sam	$200 + 400/6 = 267$	16.7%	50
Tom	$600 + 400/2 = 800$	50.0%	150
TOTAL	$1200 + 400 = 1600$	100%	300

and the new exchange rate is $1600/300 = 5.33$ \$SUSHIs for 1 \$xSUSHI. Now, if another staker Uta deposits 200 \$SUSHIs in the pool, she will receive $200/5.33 = 37.50$ \$xSUSHIs:

Stakers	\$SUSHIs	Share	\$xSUSHIs
Rose	533	29.6%	100.00
Sam	267	14.8%	50.00
Tom	800	44.4%	150.00
Uta	200	11.1%	37.50
TOTAL	1800	100%	337.50

Because of the newcomer, previous stakers see their shares of the pool reduced pro-rata, however their stakes are not diluted as they represent an unchanged quantity of sushis and the exchange rate remained at $1800/337.52 = 5.33$.

Appendix G1: Illuvium

G1.1. Game Description

Illuvium is a space exploration game that features a faraway hospitable planet hosting colorful creatures called Illuvials. In the beginning, the player's character is a pilot from an intergalactic space fleet who survived a crash on this planet during a mission investigating a distress beacon. As s/he explores the environment, the player discovers the Illuvials and their strange powers. S/he then becomes a hunter and travels across the land to *capture* Illuvials, make them *fight*, and *trade* them. Ultimately, a player must unlock "obelisks" keeping parts of the world closed off in order to continue the adventure.

G1.1.1. Illuvials

Illuvials are NFTs who are pitted against each other in battles taking place on a giant chess board with hexagonal tiles. Once creatures are installed on the board, they fight automatically without the owners' intervention. This type of game is called "autobattler". Creatures can be captured with "shards", sort of weapons that are earned or purchased.

There are over 100 different species of Illuvials, each containing many individuals. Some species are rare, others are much easier to find. Species are characterized by unique combinations of properties and statistical attributes. For example, the dog-like Dash creature can transform itself into vapor and has a high attack speed (Fig. 1). In particular, the game distinguishes "synergies" or "affinities", which concern the ability to collaborate during battles, and "fusion", a type of triad breeding by which three Illuvials are merged into a new kind (but then disappear, unlike parents who coexist with their offspring).

Affinities are based on five elements: water, fire, earth, air, and nature. There are also five classes of creature skills: fighter, guardian, rogue, psion, and empath. "Ascended affinities" are formed by the sum of two affinities, e.g.: inferno = fire + fire, frost = air + water. Similarly, "ascended classes" result from two skills combined, e.g.: slayer = fighter + rogue, invoker = psion + psion.



GI.1.2. Gather your team

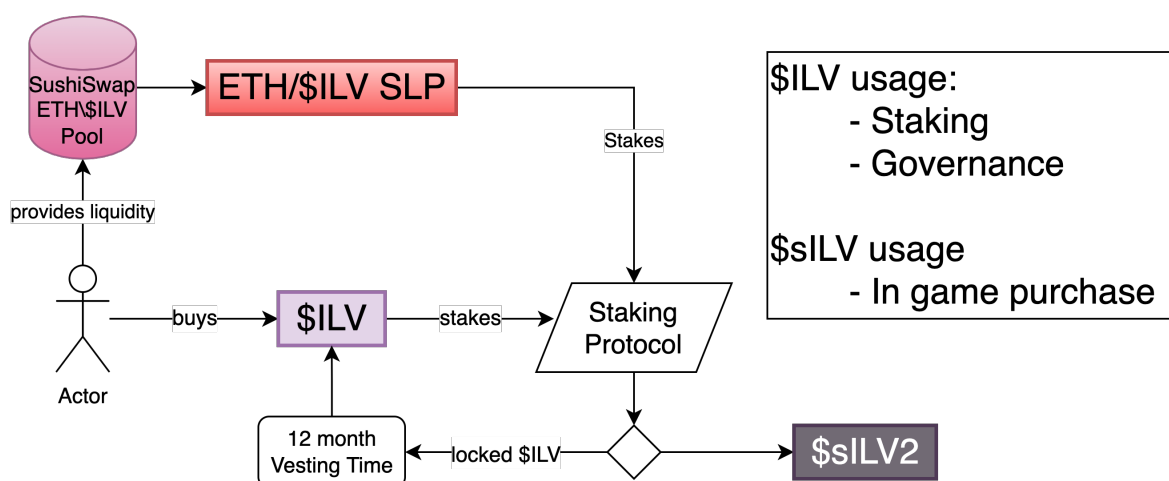
Illuvials are hard to track. They can be stored in "shards" (similar to Pokeballs). Shards have different power levels, with stronger shards used to capture stronger creatures. When an Illuvial is KO, it can regenerate over time or the owner can pay a small fee to bring it back to full health faster.

Illuvials can accumulate experience points (XPs) and climb levels. They can reach a given threshold level and if a player possesses three Illuvials of the same species at this level, they can "fuse" them into one single and more powerful Illuvial as mentioned above. Within each species, a few exceptional individuals are rarer than others, as they possess special visible traits such as "shiny", "gold", or "holographic".

Players will be able to mine free shards that are only capable of capturing a select few Illuvials that are far less powerful than some. They will be able to get started in the game without any expense and learn the mechanics.

GI.2. Tokenomics

Like Axie and Star Atlas, Illuvium relies on a dual token economics: a *governance token*, here \$ILV, typically following some issuance distribution piechart and unlock schedule, and a *game token*, here \$sILV. While \$ILV has a market value, the main difference with the other games is that \$sILV does not. Here game tokens are minted on demand. There is a free version but in order to progress, one has to buy NFTs and \$ILVs and stake them to obtain \$sILVs.



GI.2.1. Payments and treasury

Once the game is launched, players will be able to purchase various items and services, such as:

- “shard curing”
- “imbues” (modification of elements of the game such as weapons or drones)
- wagering battles
- Illuvial revival (medical fees)
- Obelisk travel
- enhancements (powerful stat boosts to the Illuvials)

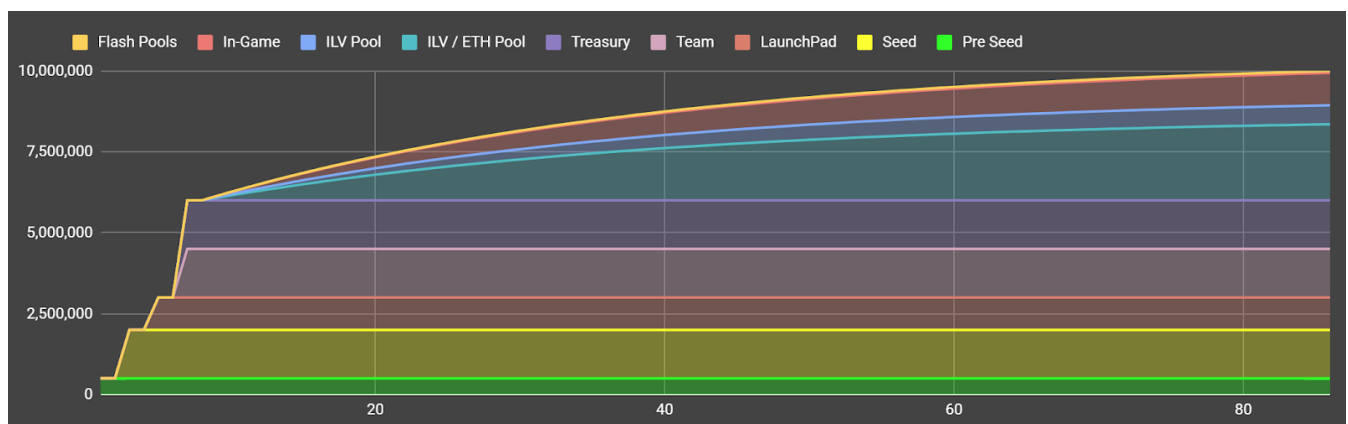
All in-game fees must be paid in ETH. These Ethers are stored into a central treasury called “The Vault”.

Players also spend money on the IlluviDEX, Illuvium’s decentralized exchange, where they can trade ETH for items that other players own. A 5% commission on the value of each transaction is applied and transferred to the Vault.

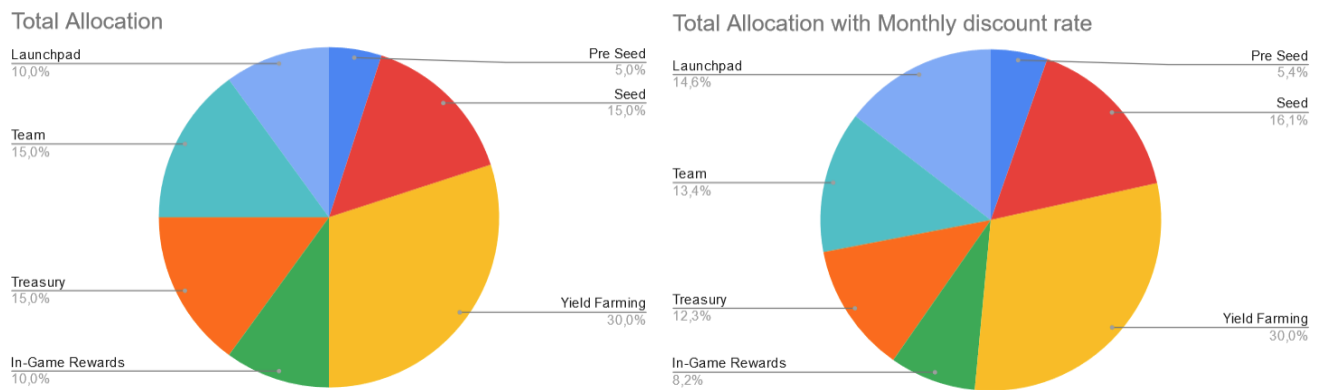
GI.2.2. Governance token

The Vault randomly buys \$ILV from the ETH/\$ILV pool on SushiSwap and distribute them to the token holders.

Distribution	Status	Amount	Month
Pre Seed	Concluded	500,000	Dec
Seed	Concluded	1,500,000	Jan
Team	Concluded	1,500,000	Feb
Treasury	Concluded	1,500,000	Feb
Launchpad	Concluded	1,000,000	Mar
Yield farming	On going	3,000,000	May
In-game rewards	Not started	1,000,000	TBC



Token distribution without and with discounted flows (see methodology in Appendix)



Compared to the undiscounted distribution, the discounted distribution diagram shows a large proportion given to launchpad (14.6% vs 10%) and Seed (16% vs. 15%). Team (13.4% vs 15%) and in-game rewards (12.3% vs. 15%) see their proportion reduced.

GI.2.3. Staking

Two Staking pools:

- \$ILV staking pool
- \$ETH/\$ILV SLP staking pool

Pool weight: Value of one pool compared to another in the case of Liquidity Mining.

\$ILV Staking Pool

- Stake \$ILV, claim \$ILV and \$sILV2.
- poolweight = 0.2

\$ETH/\$ILV Staking Pool

- Stake \$ETH/\$ILV SLP (SushiSwap Liquidity Provider token), Claim \$ILV and \$sILV2.
- poolweight = 0.8

GI.2.4. Miscellaneous observations

- Players must migrate their Ethers to Immutable X L2, where they trade NFTs. On Ethereum, one can buy ILVs. ILVs can be staked (blocked for 1 year) to get sILVs. SLPs coming from the Sushi ETH/ILV pool can also be staked. sILVs are not meant to have value outside of the game. Rewards are immediately transferred to Immutable.
- What's a little unusual is that players must buy ILVs and stake them one whole year to get sILVs which are necessary to enter the game. Therefore if the ILV rate is too high, it becomes a barrier to entrance.

- sLLVs are minted on the fly, but are they burnt? sLLVs are inflationist. The white paper is clear about the fact that sLLVs do not have any monetary value. Unlike Axie, where the game token SLP is minted upon battles and burned upon breeding, in Illuvium the sLLV is minted upon staking (off-game) then used to buy NFTs. This kind of tokenomics gives a big advantage to the team and investors.
- So how do players make money? Only by selling their creature NFTs. This is a main point of difference with Axie: here, players earn ERC-721's only, instead of ERC-20's, a game economics that could be referred to as "play to collect" (then earn).
- Also, here NFTs are burned when "fusing": three disappear for one being born, whereas Axies never disappear.

Appendix GS: Star Atlas

GS.1. Game Description

Star Atlas is a complex and ambitious game, with two white papers totalling 80 pages. Its vision is to become a “self-sovereign MMO” that is fully decentralized and sustainable(*). Compared to simpler battle games, Star Atlas is an entirely different category as it aims to create whole civilizations with an economic system featuring a vast network of multiple trades, roles, territories, and so on. To oversee this universe, it also wants to establish a DAO that would practically play the role of a government and central bank. The main aspects of the game are briefly summarized below.

(*) Eco, a centralized non-blockchain, educative game is similar in spirit:
<https://play.eco/> (<https://play.eco/>)

GS.1.1. Player progression

Player progression is based on a “**skill-tree**”: a hierarchy of customizations that players can make to their characters. It allows them to focus on certain activities from a diverse selection.

GS.1.2. Game economics

There are **three layers of economic activities** (see numbered paragraphs below):

- *A. Production*: (a) mining & refining, (b) tech crafting & blueprinting.
- *B. Piloting*: (c) exploration, (d) combat, equipment drops.
- *C. Overarching*: (e) land development, governance, character development, etc.

Most **in-game items** come from the production layer:

- *Value Tokens* (see Section “Tokenomics”):
 - the \$ATLAS token: inflationary game currency;
 - the \$POLIS token: revenue-bearing & governance token.
- *Items*: ships, crews, equipment, components, buildings
- *Real Estate*: land parcels (on planets from different solar systems).
- *Resources*: raw materials, refined materials.

(a) **Resource Management** (from *A. Production*: mining & refining)

- *Miners* yield resources used to trade or create in-game assets. Their productivity progresses based on skill (e.g., mining drill operator).
- *Mining roles* examples: extractor, refiner, manager.

(b) **Building** (from *A. Production*: tech crafting & blueprinting)

- *Builders* produce new innovative tech goods used in the game. Building is about finding the most efficient way to use farmed resources.
- *Building roles* examples: research & development, production.

(c) **Exploring** (from *B. Piloting*: exploration)

- *Explorers* use ships and tech to bring back materials for miners and builders. They also depend on fighters and builders to minimize mission risk.
- *Exploring roles* examples: executive roles, utility roles.

(d) **Fighting** (from *B. Piloting*: combat)

- *Fighters* capture loot (resources and tech) from other players. They depend on builders to gain advantages in combat.
- *Fighting roles* examples: executive roles, combat roles.

(e) **Overarching Activities** (from *C. Overarching*)

- *Other secondary roles* that make Star Atlas more fun, while adding value to the economy.
- *Overarching roles* examples: retail deck manufacturers/sellers, social developers, sporting activities.

GS.2. Tokenomics

Star Atlas has two fungible tokens that fuel all operations, **\$POLIS** and **\$ATLAS**:

- *\$POLIS is the governance token* representing financial stake in the game, voting power in the DAOs, and control of the Treasury. It has a fixed total supply of 360M (*unless voted otherwise*).
- *\$ATLAS is the in-game payment token*. It is inflationary, to match the growth of the economy and “provide a solid monetary base”. It also comes in limited quantity, here 36B.

As in Axie and Illuvium, Star Atlas rests on a dual-token economy. Unlike Illuvium's \$sILV, however, the \$ATLAS game token has monetary value and it is also meant to be traded on secondary markets. Interestingly, \$ATLAS also obeys a total supply and is distributed like the governance token \$POLIS, in particular to investors. It is doubtful that they are going to use it to play the game, though, so they will probably wait until it acquires value to resell it.

GS.2.1. Governance token

\$POLIS can be acquired in the following ways:

- *Primary issuance*: the team directly sells POLIS tokens to potential governors
- *Secondary markets*: can be traded under a finite variety of lockup/vesting schedules
- *In-game rewards*: upon completing certain missions
- *Metagame rewards*: to actively attract new users staying long enough; also organize users into guilds/factions rewarded by participation
- *Governance participation*: Staking in the DAO and being active
- *Meta-economics rewards*: Liquidity provision on AMMs and other DeFi entities

The total supply of \$POLIS is 360M tokens which will be distributed as in the table below.

GS.2.2. Game token

\$ATLAS is the in-game currency earned within the game.

Via the investment cycle, players should end up with more \$ATLAS at the end of the game than they had in the beginning. All games have a "utility loop" of this kind, which gives players a net gain in game tokens. Here this is accomplished by an *inflationist \$ATLAS issuance*, which is regulated as a function of the number of players, and in such a way that its mass grows at the same rate as the game economy.

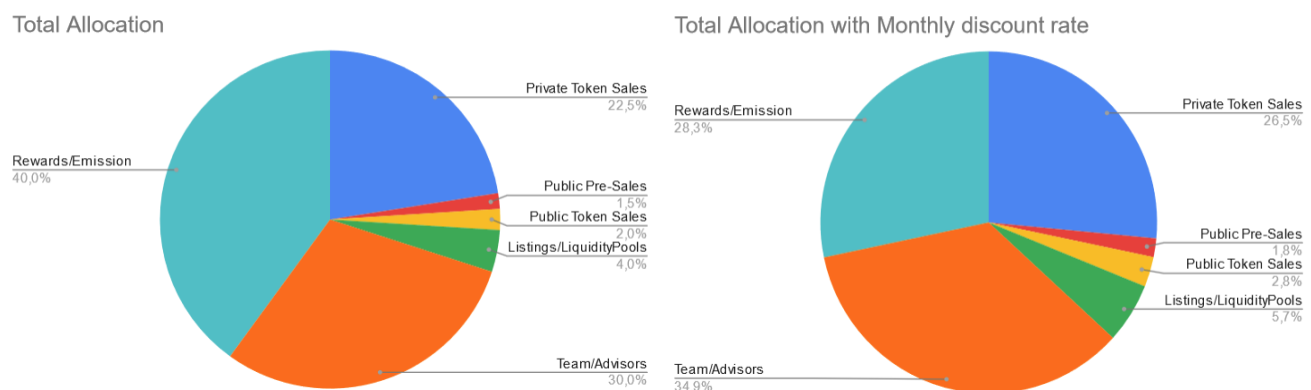
A few principles guiding \$ATLAS issuance:

- New \$ATLAS tokens are minted to reward in-game activities. These rewards can be parametrized by the \$POLIS DAO.
- Minting and burning of \$ATLAS is purely based on game user-base growth and adoption.
- Supply and price are kept on a level to maintain reasonable in-game asset pricing and optimize entry level for new users.

The total supply of \$ATLAS is 36B tokens which will be distributed as follows:

Recipient Group	\$POLIS Allocation (360M)	\$ATLAS Allocation (36B)	Vesting Period
Private token sales	22.5% (81M)	22.5% (8.1B)	2 years
Public pre-sale	1.5% (5.4M)	1.5% (540M)	2 years
Public token sales	2% (7.2M)	2% (720M)	0
Liquidity pools	4% (14.4M)	4% (1.44B)	0
Rewards/emission	40% (144M)	65% (23.4B)	8 years
Team	30% (108M)	5% (1.8B)	0 or 2 years

Token distribution without and with discounted flows (see methodology in Appendix)



Compared to the undiscounted distribution, the discounted distribution diagram shows a larger proportion given to private token sales (26.5% vs 22.5%) and Team/advisors (34.9% vs. 30%). Rewards emission (28.3% vs. 40%) sees its proportion greatly reduced.

The **monetary policy of \$ATLAS** will follow three phases:

1. *Discretionary (central bank)*: the team adjusts the emission rates directly, using its DAO votes to modify multipliers on in-game rewards.
2. *Discretionary (decentralized governance)*: emission rates are still based on modifying reward multipliers, but team votes are phased out, and the DAO takes control.
3. *Parametric (decentralized governance)*: the system is replaced with an algorithmic stabilization model to set targets for the monetary mass, and adjust minting/burning accordingly.

In sum, Star Atlas's core tokenomics loop flows between in-game fees and in-game rewards. A fixed % of the collected fees fills the Rewards Fund; the rest is sent into the DAO Treasury. The DAO can subsidize the Fund and/or adjust monetary parameters (inflation target; monetary mass by burning). It can also balance the in-game economy by selectively dis/incentivizing certain activities.

Generic questions one should be asking about each game:

- How does the governance token capture the game value? Via shares/stakes?
- How does the game token maintain a relatively stable market price? Via automated mint & burn? Or via a "central bank"-type DAO that enforces a true monetary policy, e.g. with an inflation target?

GS.2.3. NFTs

Star Atlas also manages **non-monetary assets**:

- Main categories: ships, equipment, components, crew.
- Every item is an NFT recorded on the blockchain.
- NFTs reflect the ownership, specifics, and rarity of the items.

- NFTs can be purchased in-game using \$ATLAS or on marketplaces using Solana-supported currencies.
- Items can be either dropped (e.g., via quests) or crafted (by combining necessary resources).
- Items can also be improved to increase their rarity.

GS.2.4. Land

- Planets are divided into land parcels recorded as on-chain NFTs.
- Buildings are built on land.
- Land is owned rather than rented, protecting users from eviction or discrimination.
- To disincentivize "flipping", the game utilizes Land Value Tax (LVT), which is priced by the DAO.
- If the owner does not pay taxes, the land is foreclosed, disabling all buildings on it, then auctioned off.

- Land can be acquired for stablecoins or \$ATLAS: whether in auctions, or from parcels issued by the team, or from existing land owners who are selling.
- Land owners can be incentivized through \$POLIS (depending on their behavior) or \$ATLAS (depending on their productive activities)
- They may use their parcels as collateral for loans in \$ATLAS.

/!\ "Land-like goods"

What is land? It is not a product, but an ownership share of three-dimensional space

<https://www.fresheconomicthinking.com/2018/09/what-is-land-it-is-not-product-but.html> (<https://www.fresheconomicthinking.com/2018/09/what-is-land-it-is-not-product-but.html>)

GS.2.5 Additional DeFi functionalities

The above list of \$ATLAS, \$POLIS and NFT functions are non-exhaustive and subject to further changes:

- Some of the tokens' emission will be allocated to staking and yield farming.
- Their use may extend beyond the game into the broader DeFi space within the Solana ecosystem.
- For instance, any token may be listed under any number of DEXes and used for lending, collaterals, and yield farming.

GS.3. Governance

*"The long-term vision of Star Atlas is an autonomous, player-owned game, fully controlled and funded by decentralized governance. Therefore, from its early stages, the economy is architected to run through a **hierarchy of DAOs**, fueled by the governance token POLIS and cash flows of ATLAS."* (Economics white paper, p28)

"The DAO is designed to allow the investors and players to take over the reins eventually, yet still allow a reasonable degree of control for the developers in the beginning." (Economics white paper, p34)

The DAO is organized in multiple levels, each requiring a stake in \$POLIS tokens. The general idea is that, going top-down, requirements of in-game participation increase, while staking decreases.

(a) **The top-level governance** regulates the overall well-being of the game:

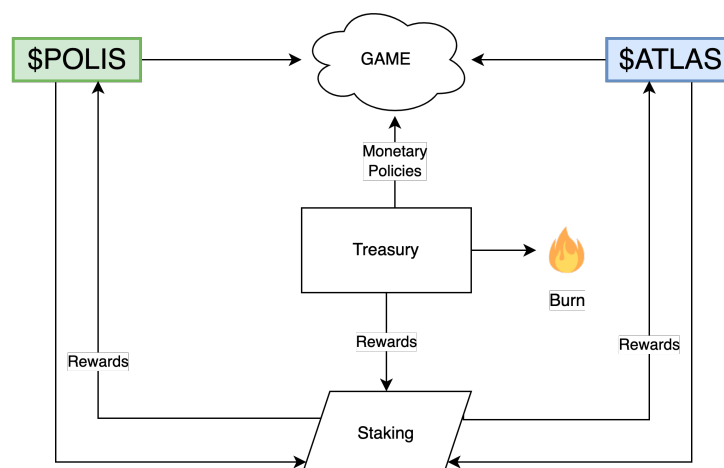
- The DAO votes on developments to lend, grant, and invest in new developments.
- DAO members also earn portions of the tax revenue.
- The DAO defines: a monetary policy; the Treasury (for game development, NPC purchases, dividends); the rules of lower-level DAOs; and misc aspects (emergency actions, disabling game items).

(b) **Faction DAOs**, the middle-level governance, organizes players at the scale of large competing groups:

- They correspond to three core territories.
- Faction DAOs set faction-wide targets: occupy, hold, build, patrol.
- Targets require staking \$POLIS.

(c) **Regional DAOs**, the low-level governance, control star systems by adjusting their access and taxation on activities:

- Voters in regional DAOs must own property.
- Taxation can be done by a flat % on all activities or specialized by activity.
- Regional treasuries cannot pay dividends, but can provide monetary bonuses for certain missions.
- Regional DAOs also define a military policy (protecting the planets), an internal security (protecting the homeland), and a diplomacy (making alliances, trade agreements).



Appendix GX: Axie

GX.1. Game

Axie Infinity (<https://axieinfinity.com> (<https://axieinfinity.com>)) is a breeding & battle game based on an original collection of NFTs representing small blob-like creatures. The original principle of this game was limited to *breeding*, whereby two creatures could mate to produce a third one. This mating process, still active today, consumes two ERC-20 tokens: the \$AXS (Axie Infinity Shards) and the \$SLP (Smooth Love Potion). In a second phase of deployment, the Axie team released a one-on-one, turn-based *battle* card game. A battle consists of two teams of three Axies, and ends when all three Axies from one team are knocked out. This battle card game is the one that generates the most \$SLP tokens, since battle winners earn \$SLPs.

GX1.1. Game mechanics

The card deck is based on the three Axies chosen by the player to be part of their team. Each Axie is unique and has specific “body parts” inherited from its parents. Each body part corresponds to one card of a deck: for example, if an Axie has fangs, it corresponds to a “bite” card.

The goal of the game is to constitute a strong deck where body parts are in good match and “collaborate” to maximize the odds of winning the game. One has to be careful that each Axie has a specific “type” among 8 possible types (similar to species), hence breeding cannot happen between any two Axies. Inspired by real-world genetics, each Axie possesses “dominant” and “recessive” alleles, which have a greater or lesser likelihood to be transmitted to their offspring. Therefore, since breeding has a cost, it would be a waste to attempt any random breeding.

GX1.2. Tokenomics

There are two main ERC-20 tokens: \$AXS and \$SLP, and the collection of ERC-721 NFTs. \$AXS is the main token of the Axie ecosystem. Breeding costs 1 \$AXS + a certain number of \$SLPs which depends on how many children were already created by the Axie: 900 \$SLPs if it had no children, 1350 if it already had 1 child, and so on, until 15,300 \$SLPs if it had 6 children. This exponential law of \$SLP cost was designed to avoid a demographic explosion or “hyperinflation” of the Axie NFTs.

\$AXS can also be staked in the “Community Treasury”, which grows as the game collects breeding fees. \$AXS stakers earn a high APR interest rate, typically between 35%-70%. According to the light paper, the \$AXS token is also envisioned to become a governance

token in a future phase of the game.

Axie NFTs can be traded on the marketplace. They are priced in Ethers and the Axie Infinity protocol takes a 5.25% fee on each sale, which is deposited into the Community Treasury.

There is a high bar to earn \$AXS tokens: only when players reach a high rank in the world leader-board, i.e. climbs up the "ladder", can they claim their \$AXS gains. For example, the first 50 players share a total of 200 \$AXS.

The \$SLPs are a more common kind of "reward" tokens: they are given to players every time they win a battle, with a daily cap of typically 50 \$SLPs. \$SLPs have only one use: breeding. They can also be traded like any other ERC-20.

GX.2. Allocation

According to the unlock schedule and tables advertized in the white paper:

<https://whitepaper.axieinfinity.com/axs/allocations-and-unlock> (<https://whitepaper.axieinfinity.com/axs/allocations-and-unlock>)

<https://docs.google.com/spreadsheets>

[/d/1zXviXJ17_Zqze9XajYt2OhKroGA5Po_o566LFWIc1QY/edit?usp=sharing](https://docs.google.com/spreadsheets/d/1zXviXJ17_Zqze9XajYt2OhKroGA5Po_o566LFWIc1QY/edit?usp=sharing)

(https://docs.google.com/spreadsheets/d/1zXviXJ17_Zqze9XajYt2OhKroGA5Po_o566LFWIc1QY/edit?usp=sharing)

- The first version of Axie as a breeding-only game was launched in May 2018.
- The maximum total supply of \$AXS is set at 270M and should be reached by month 65 (five and a half years).
- Team Sky Mavis, the Axie founders and employees, has allocated itself over 10M \$AXS at the start, then over 5M every half-year.
- Gaming issuance has been allocated a total of 54M \$AXS and is released in decreasing fashion at a rate of about 25% every year, going from 9M in May 2021 to 6.7M in 2022, etc.
- Relatively independently from the fluctuations of the Ether, the market value of \$SLP reached a peak in July 2021 at \$0.40 while \$AXS was at a maximum in November 2021 at \$64.
- These peaks reveal a huge increase in players' interest, thus a scarcity in \$SLPs caused by greater breeding numbers.

GX.2.1. Main points

In sum, the Axie's game designers paired up two tokens:

- one token with an increasing supply (\$SLP), whose role is to facilitate and encourage

game flows;

- another token whose maximum supply is capped and accrues value over time with game adoption (\$AXS).

The rationale of this pairing could be to “stabilize” the game, in a way similar to gold-silver “bimetallism” in economics. The usefulness of this bi-token system should be confirmed by its ability to avoid the pitfall of a single-token game that can either collapse if the token loses its value or, on the contrary, block newcomers if it becomes too expensive.

On the whole, however, it appears that the Axie ecosystem suffered from a “bubble effect”, i.e. a sudden growth in popularity and a too rapid expansion of the players’ community. This was logically followed by a correction and significant drop in the \$AXS token valuation and game frequentation.

Another major problem is that Axie is missing a mechanism of NFT burning. This leads to an overpopulation of creatures, many of them are worth nothing. By contrast, Illuvium (another game) discards three creatures everytime they fuse into a new one, hence maintaining NFT scarcity and value.

The evolution of the number of players is correlated with the number of Axies bred and \$SLP tokens minted. The more players, the more new Axie NFTs are needed, hence the more expensive they become. Yet again, the reward token \$SLP went through bouts of hyperinflation due to the fact that it does not have much use beside breeding, and breeding was losing its usefulness and attractiveness as the number of newborn Axies exploded.

